

317521: Computer Networks

Unit Number: 5

Unit Name: Application Layer

Unit Outcomes: CO4

CO4: Implement client-server applications using sockets

Syllabus

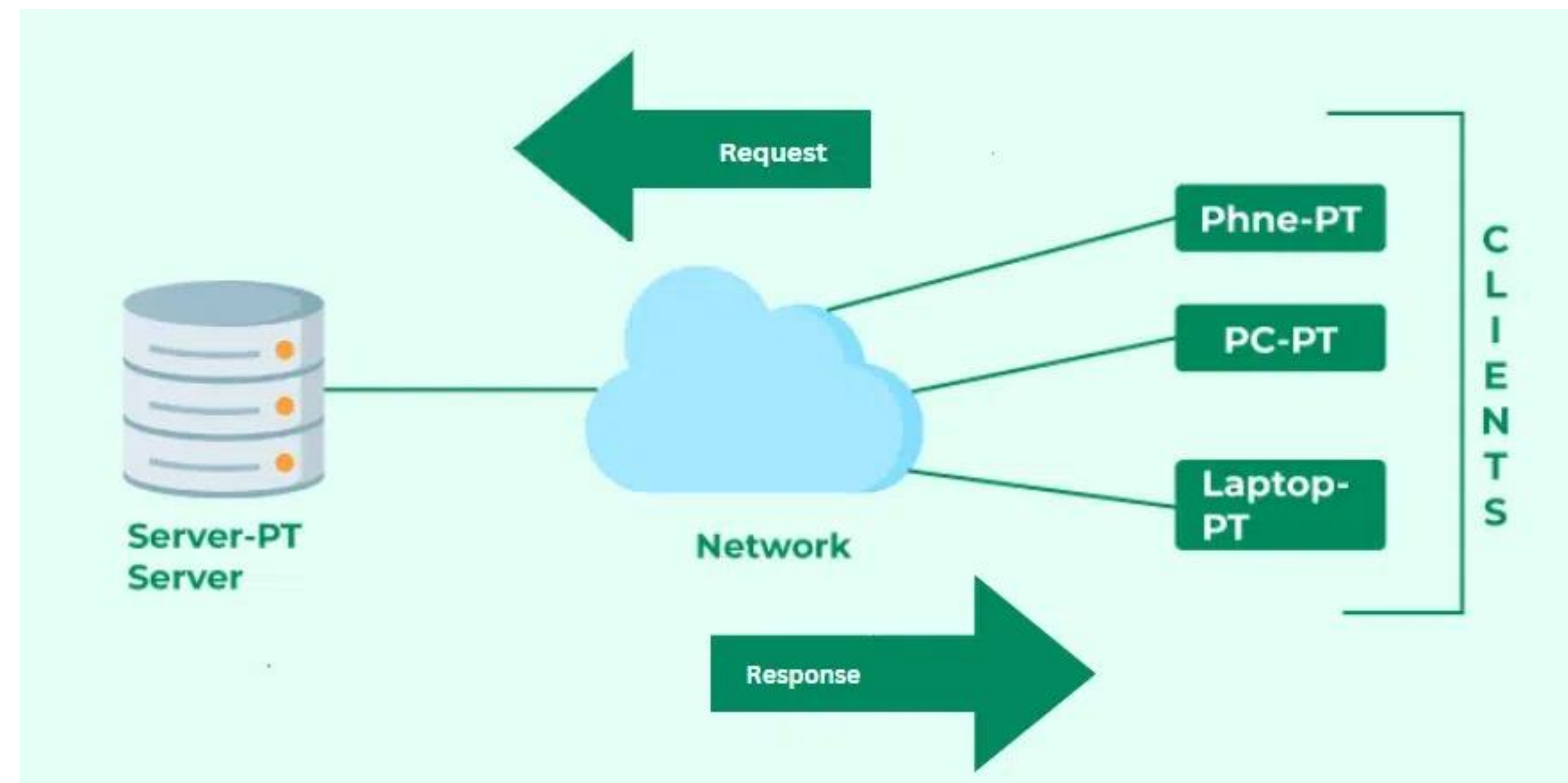
- **Client Server Paradigm, Peer to Peer Paradigm**
- **Communication using TCP and UDP services, Domain Name System (DNS), HyperText Transfer Protocol (HTTP)**
- **Email: SMTP, MIME, POP3, Webmail, FTP, TELNET**
- **Dynamic Host Control Protocol (DHCP), Simple Network Management Protocol (SNMP).**

Client-Server Model:

- The Client-server model is a distributed application structure that partitions tasks or workloads between the providers of a resource or service, called servers, and service requesters called clients.
- In the client-server architecture, when the client computer sends a request for data to the server through the internet, the server accepts the requested process and delivers the data packets requested back to the client.
- Clients do not share any of their resources. Examples of the Client-Server Model are Email, World Wide Web, etc.

How Does the Client-Server Model Work:

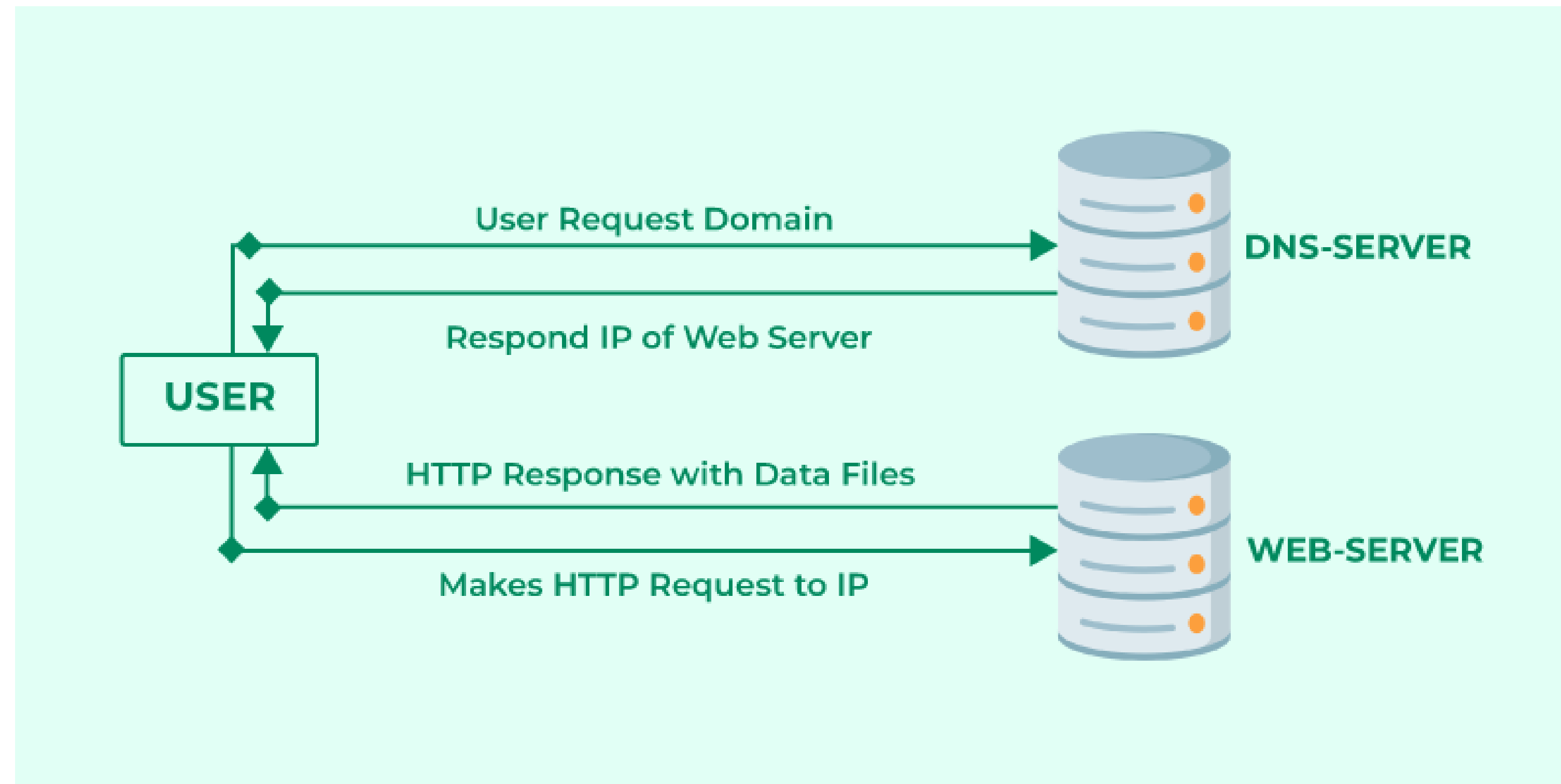
- . **Client:** When we say the word **Client**, it means to talk of a person or an organization using a particular service. Similarly in the digital world, a **Client** is a computer (**Host**) i.e. capable of receiving information or using a particular service from the service providers (**Servers**).
- . **Servers:** Similarly, when we talk about the word **Servers**, It means a person or medium that serves something. Similarly in this digital world, a **Server** is a remote computer that provides information (data) or access to particular services.



How the Browser Interacts With the Servers:

There are a few steps to follow to interact with the servers of a client.

- . User enters the **URL** (Uniform Resource Locator) of the website or file. The Browser then requests the **DNS** (DOMAIN NAME SYSTEM) Server.
- . **DNS Server** lookup for the address of the **WEB Server**.
- . The **DNS Server** responds with the **IP address** of the **WEB Server**.
- . The Browser sends over an **HTTP/HTTPS** request to the **WEB Server's IP** (provided by the **DNS server**).
- . The Server sends over the necessary files for the website.
- . The Browser then renders the files and the website is displayed. This rendering is done with the help of **DOM** (Document Object Model) interpreter, **CSS** interpreter, and **JS Engine** collectively known as the **JIT** or (Just in Time) Compilers.



Advantages of Client-Server Model

- . Centralized system with all data in a single place.
- . Cost efficient requires less maintenance cost and Data recovery is possible.
- . The capacity of the Client and Servers can be changed separately.

Disadvantages of Client-Server Model

- . Clients are prone to viruses, Trojans, and worms if present in the Server or uploaded into the Server.
- . Servers are prone to Denial of Service (DOS) attacks.
- . Data packets may be spoofed or modified during transmission.
- . Phishing or capturing login credentials or other useful information of the user are common and MITM (Man in the Middle) attacks are common.

What is P2P (Peer-to-Peer Process)?

- A peer-to-peer network is a simple network of computers. Here each computer acts as a node for file sharing within the formed network. Here each node acts as a server and thus there is no central server in the network.
- This allows the sharing of a huge amount of data. The tasks are equally divided amongst the nodes.
- Each node connected in the network shares an equal workload. For the network to stop working, all the nodes need to individually stop working. This is because each node works independently

Types of P2P networks:

Unstructured P2P networks: In this type of P2P network, each device is able to make an equal contribution. This network is easy to build as devices can be connected randomly in the network. But being unstructured, it becomes difficult to find content. For example, Napster, Gnutella, etc.

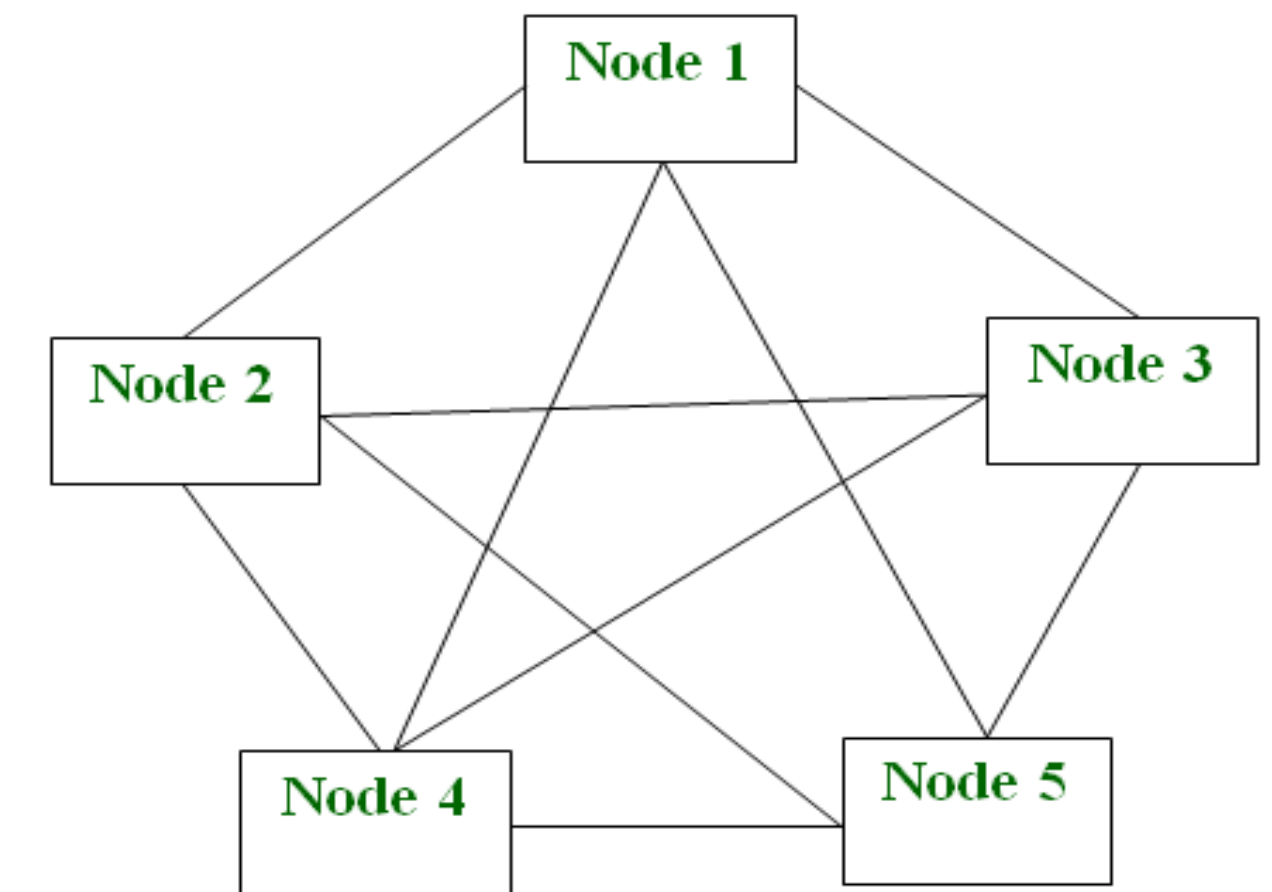
Structured P2P networks: It is designed using software that creates a virtual layer in order to put the nodes in a specific structure. These are not easy to set up but can give easy access to users to the content. For example, P-Grid, Kademlia, etc.

Hybrid P2P networks: It combines the features of both P2P networks and client-server architecture. An example of such a network is to find a node using the central server.

P2P Network Architecture

In the P2P network architecture, the computers connect with each other in a workgroup to share files, and access to internet and printers.

- Each computer in the network has the same set of responsibilities and capabilities.
- Each device in the network serves as both a client and server.
- The architecture is useful in residential areas, small offices, or small companies where each computer act as an independent workstation and stores the data on its hard drive.
- Each computer in the network has the ability to share data with other computers in the network.
- The architecture is usually composed of workgroups of 12 or more computers.



P2P Architecture

Applications of P2P Network

Below are some of the common uses of P2P network:

- . **File sharing:** P2P network is the most convenient, cost-efficient method for file sharing for businesses. Using this type of network there is no need for intermediate servers to transfer the file.
- . **Blockchain:** The P2P architecture is based on the concept of decentralization. When a peer-to-peer network is enabled on the blockchain it helps in the maintenance of a complete replica of the records ensuring the accuracy of the data at the same time. At the same time, peer-to-peer networks ensure security also.
- . **Direct messaging:** P2P network provides a secure, quick, and efficient way to communicate. This is possible due to the use of encryption at both the peers and access to easy messaging tools.
- . **File sharing networks:** Many P2P file sharing networks like G2, and eDonkey have popularized peer-to-peer technologies.
- . **IP Telephony:** Skype is one good example of a P2P application in VoIP.

Advantages of P2P Network:

- . **Easy to maintain:** The network is easy to maintain because each node is independent of the other.
- . **Less costly:** Since each node acts as a server, therefore the cost of the central server is saved. Thus, there is no need to buy an expensive server.
- . **No network manager:** in a P2P network since each node manages his or her own computer, thus there is any need for a network manager.
- . **Adding nodes is easy:** Adding, deleting, and repairing nodes in this network is easy.
- . **Less network traffic:** In a P2P network, there is less network traffic than in a client/ server network.

Disadvantages of P2P Network

- . **Data is vulnerable:** Because of no central server, data is always vulnerable to getting lost because of no backup.
- . **Less secure:** It becomes difficult to secure the complete network because each node is independent.
- . **Slow performance:** In a P2P network, each computer is accessed by other computers in the network which slows down the performance of the user.
- . **Files hard to locate:** In a P2P network, the files are not centrally stored, rather they are stored on individual computers which makes it difficult to locate the files.

Domain Name System (DNS) in Application Layer:

The Domain Name System (DNS) is like the internet's phone book. It helps you find websites by translating easy-to-remember names (like `www.example.com`) into the numerical IP addresses (like `192.0.2.1`) that computers use to locate each other on the internet. Without DNS, you would have to remember long strings of numbers to visit your favourite websites.

Domain Name System (DNS) is a hostname used for **IP address** translation services. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers. It is required for the functioning of the Internet.

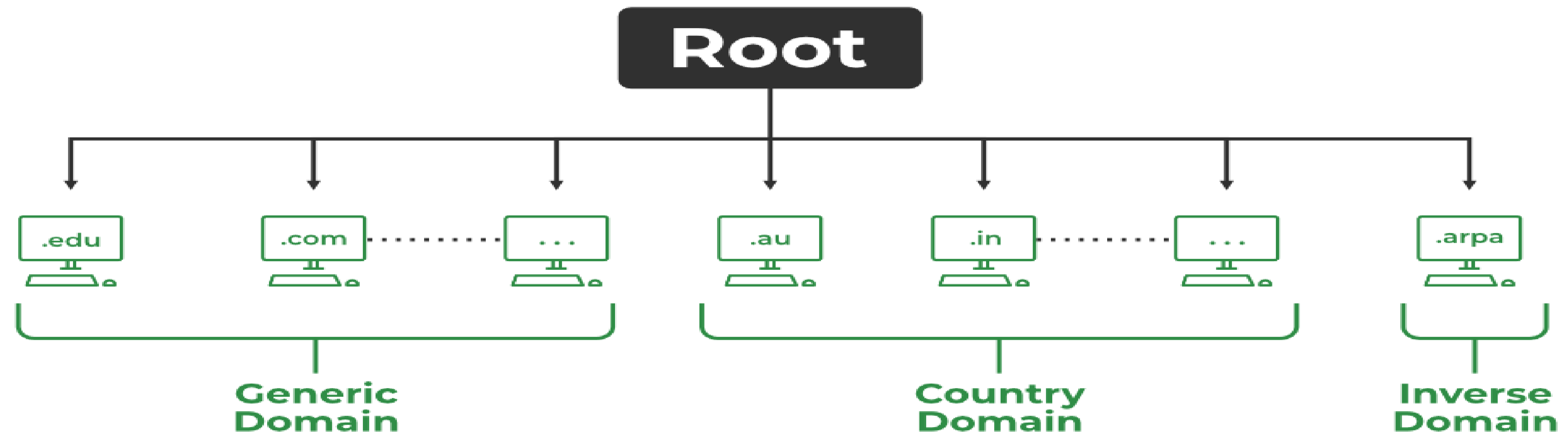
What is the Need for DNS?

Every host is identified by the IP address but remembering numbers is very difficult for people also the IP addresses are not static therefore a mapping is required to change the domain name to the IP address. So DNS is used to convert the domain name of the websites to their numerical IP address.

Types of Domain

There are various kinds of domains:

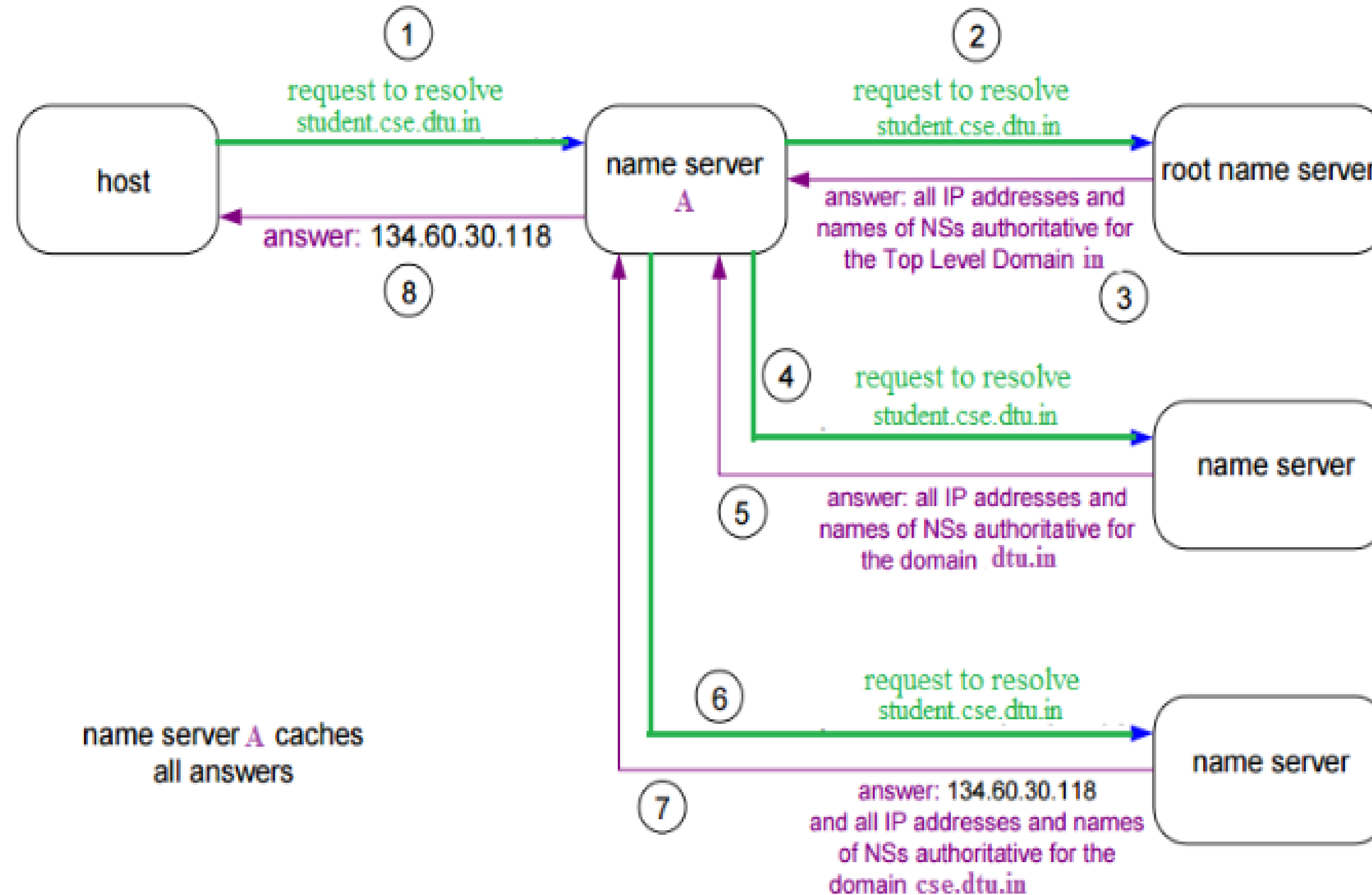
- **Generic Domains:** .com (commercial), .edu (educational), .mil (military), .org (non-profit organization), .net(similar to commercial) all these are generic domains.
- **Country Domain:** .in (India) .us .uk
- **Inverse Domain:** if we want to know what is the domain name of the website. IP to domain name mapping. So DNS can provide both the mapping for example to find the IP addresses of geeksforgeeks.org then we have to type *nslookup*



Types of DNS

Domain Name Server:

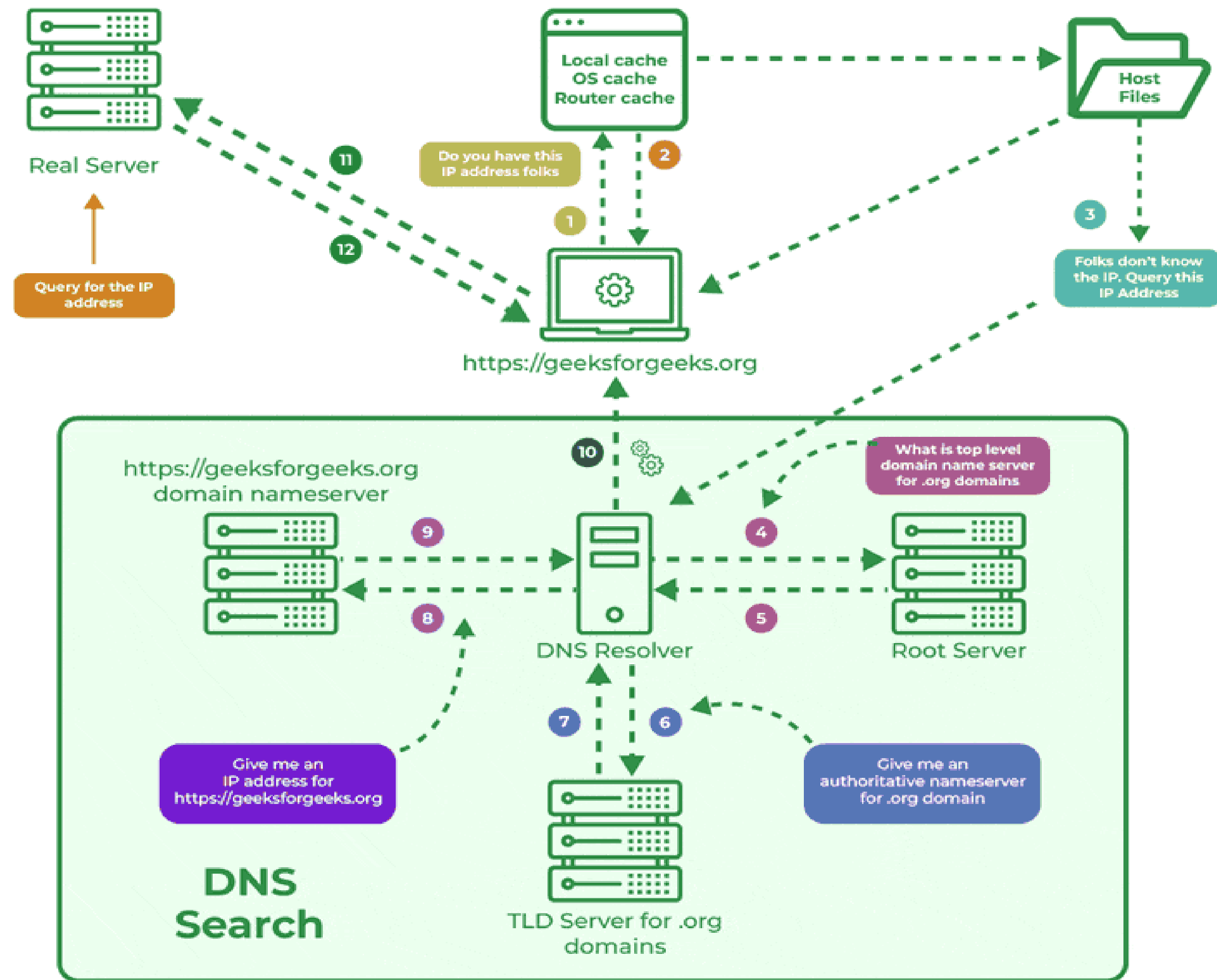
The client machine sends a request to the local name server, which, if the root does not find the address in its database, sends a request to the root name server, which in turn, will route the query to a top-level domain (TLD) or authoritative name server. The root name server can also contain some hostName to IP address mappings. The Top-level domain (TLD) server always knows who the authoritative name server is. So finally the IP address is returned to the local name server which in turn returns the IP address to the host.



How Does DNS Work:

- The working of DNS starts with converting a hostname into an IP Address. A domain name serves as a distinctive identification for a website.
- It is used in place of an IP address to make it simpler for consumers to visit websites. Domain Name System works by executing the database whose work is to store the name of hosts which are available on the Internet.
- The top-level domain server stores address information for top-level domains such as .com and .net, .org, and so on. If the Client sends the request, then the DNS resolver sends a request to DNS Server to fetch the IP Address.
- In case, when it does not contain that particular IP Address with a hostname, it forwards the request to another DNS Server. When IP Address has arrived at the resolver, it completes the request over Internet Protocol.

How Does DNS Works



HTTP:

- **HTTP (Hypertext Transfer Protocol) is a fundamental protocol** of the Internet, enabling the transfer of data between a client and a server. It is the foundation of data communication for the World Wide Web.
- HTTP provides a standard between a web browser and a web server to establish communication. It is a set of rules for transferring data from one computer to another.
- Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP.
- It is an application protocol that is used for distributed, collaborative, hypermedia information systems.

Methods of HTTP

- . **GET**: Used to retrieve data from a specified resource. It should have no side effects and is commonly used for fetching web pages, images, etc.
- . **POST**: Used to submit data to be processed by a specified resource. It is suitable for form submissions, file uploads, and creating new resources.
- . **PUT**: Used to update or create a resource on the server. It replaces the entire resource with the data provided in the request body.
- . **PATCH**: Similar to PUT but used for partial modifications to a resource. It updates specific fields of a resource rather than replacing the entire resource.

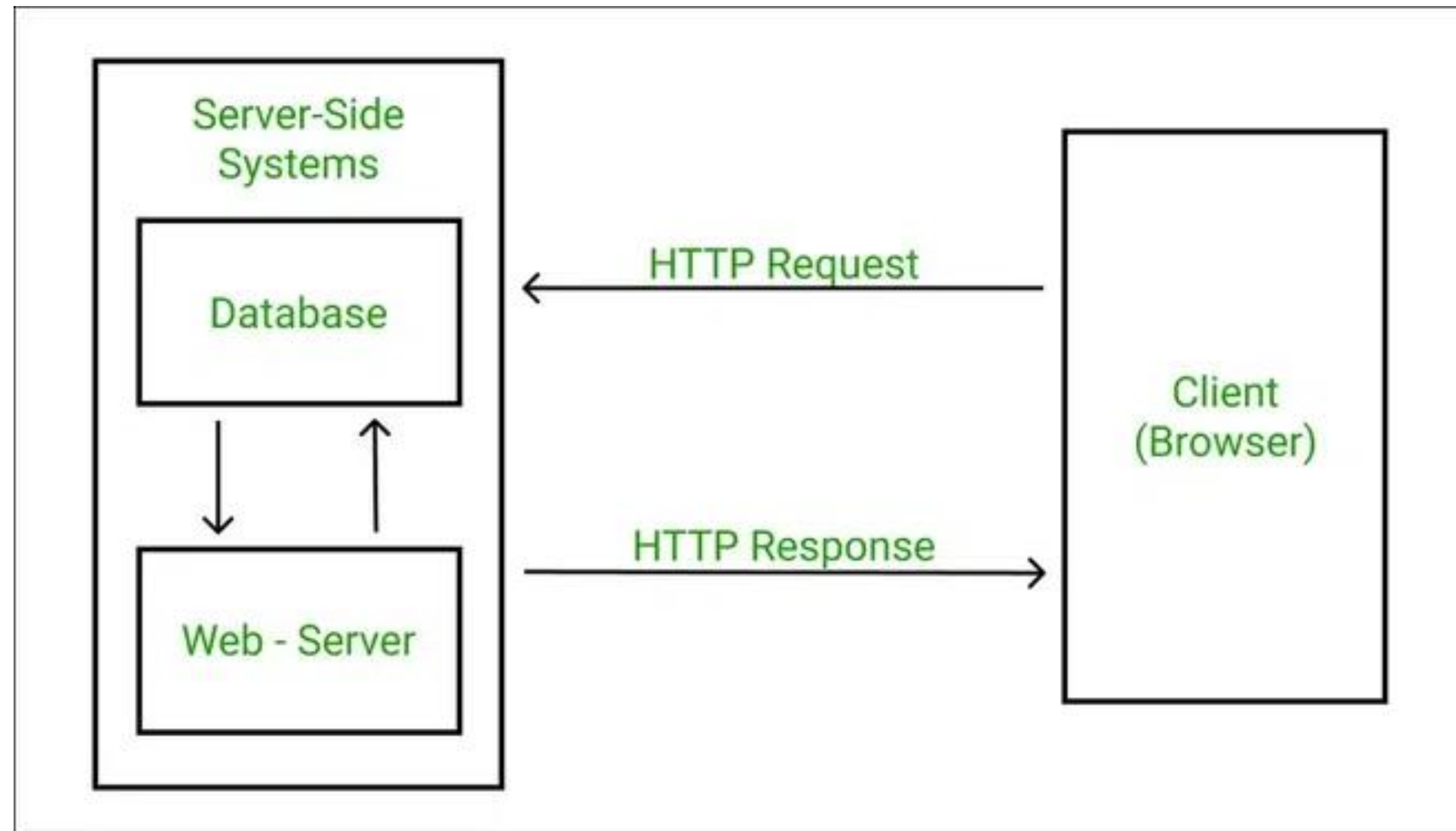
- . **DELETE**: Used to remove a specified resource from the server.
- . **HEAD**: Similar to GET but retrieves only the response headers, useful for checking resource properties without transferring the full content.
- . **OPTIONS**: Used to retrieve the communication options available for a resource, including supported methods and headers.
- . **TRACE**: Used for debugging purposes to echo the received request back to the client, though it's rarely used due to security concerns.
- . **CONNECT**: Used to establish a tunnel to the server through an HTTP proxy, commonly used for SSL/TLS connections.

HTTP Request/Response:

HTTP is a request-response protocol, which means that for every request sent by a client (typically a web browser), the server responds with a corresponding response. The basic flow of an HTTP request-response cycle is as follows:

- . **Client sends an HTTP request:** The client (usually a web browser) initiates the process by sending an HTTP request to the server. This request includes a request method (GET, POST, PUT, DELETE, etc.), the target URI (Uniform Resource Identifier, e.g., a URL), headers, and an optional request body.
- . **Server processes the request:** The server receives the request and processes it based on the requested method and resource. This may involve retrieving data from a database, executing server-side scripts, or performing other operations.
- . **Server sends an HTTP response:** After processing the request, the server sends an HTTP response back to the client. The response includes a status code (e.g., 200 OK, 404 Not Found), response headers, and an optional response body containing the requested data or content.

- **Client processes the response:** The client receives the server's response and processes it accordingly. For example, if the response contains an HTML page, the browser will render and display it. If it's an image or other media file, the browser will display or handle it appropriately.

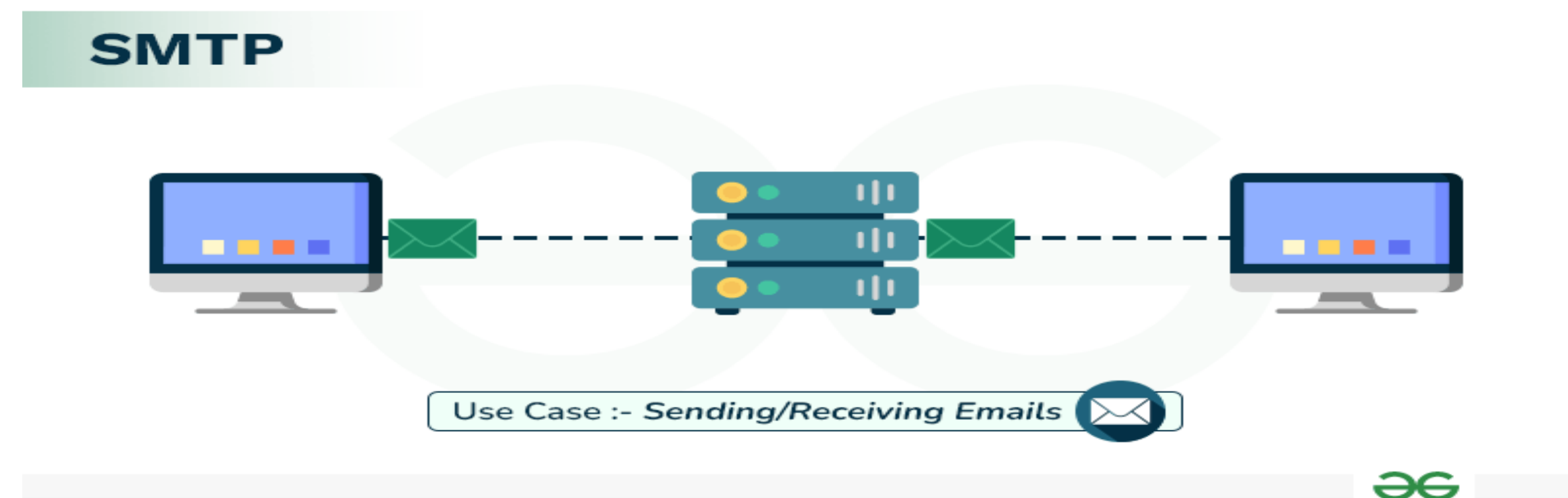


SMTP:

Simple Mail Transfer mechanism (SMTP) is a mechanism for exchanging email messages between servers. It is an essential component of the email communication process and operates at the application layer of the TCP/IP protocol stack. SMTP is a protocol for transmitting and receiving email messages. In this article, we are going to discuss every point about SMTP.

What is Simple Mail Transfer Protocol?

SMTP is an application layer protocol. The client who wants to send the mail opens a TCP connection to the SMTP server and then sends the mail across the connection. The SMTP server is an always-on listening mode. As soon as it listens for a TCP connection from any client, the SMTP process initiates a connection through port 25. After successfully establishing a TCP connection the client process sends the mail instantly.



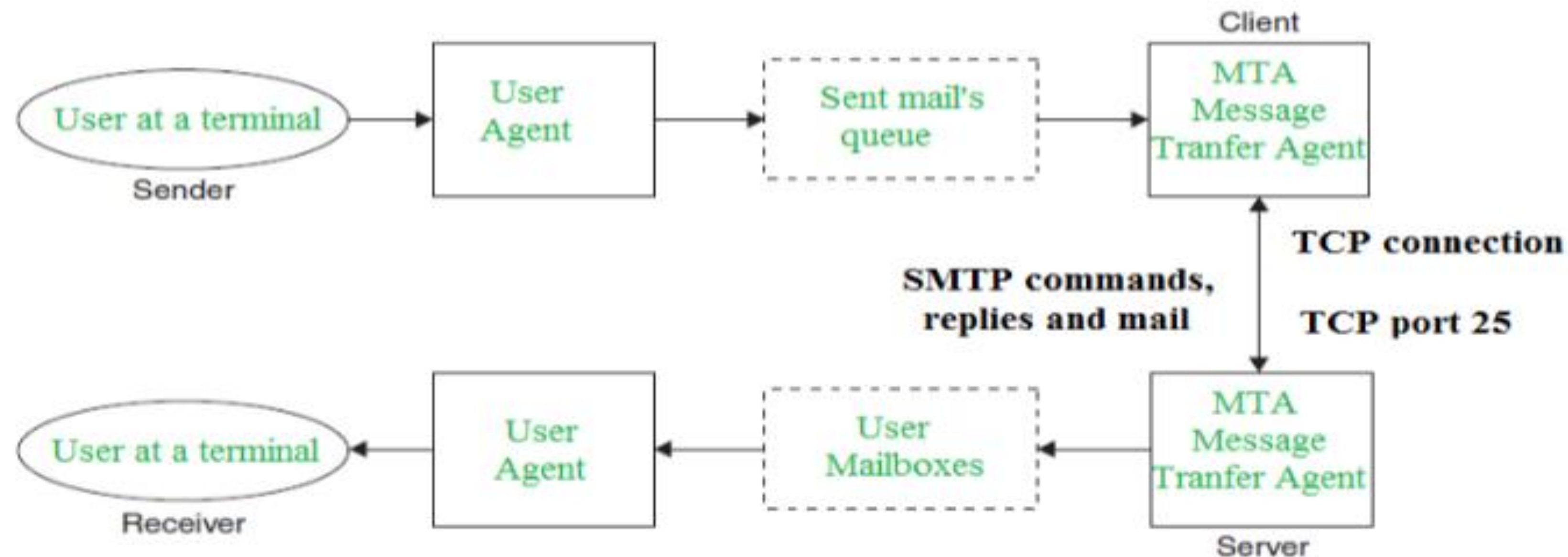
SMTP Protocol:

The SMTP model is of two types:

- . End-to-End Method
 - . Store-and-Forward Method
- The end-to-end model is used to communicate between different organizations whereas the store and forward method is used within an organization.
 - An SMTP client who wants to send the mail will contact the destination's host SMTP directly, to send the mail to the destination. The SMTP server will keep the mail to itself until it is successfully copied to the receiver's SMTP.
 - The client SMTP is the one that initiates the session so let us call it the client-SMTP and the server SMTP is the one that responds to the session request so let us call it receiver-SMTP. The client-SMTP will start the session and the receiver SMTP will respond to the request.

Model of SMTP System

In the SMTP model user deals with the user agent (UA), for example, Microsoft Outlook, Netscape, Mozilla, etc. To exchange the mail using TCP, MTA is used. The user sending the mail doesn't have to deal with MTA as it is the responsibility of the system admin to set up a local MTA. The MTA maintains a small queue of mail so that it can schedule repeat delivery of mail in case the receiver is not available. The MTA delivers the mail to the mailboxes and the information can later be downloaded by the user agents.

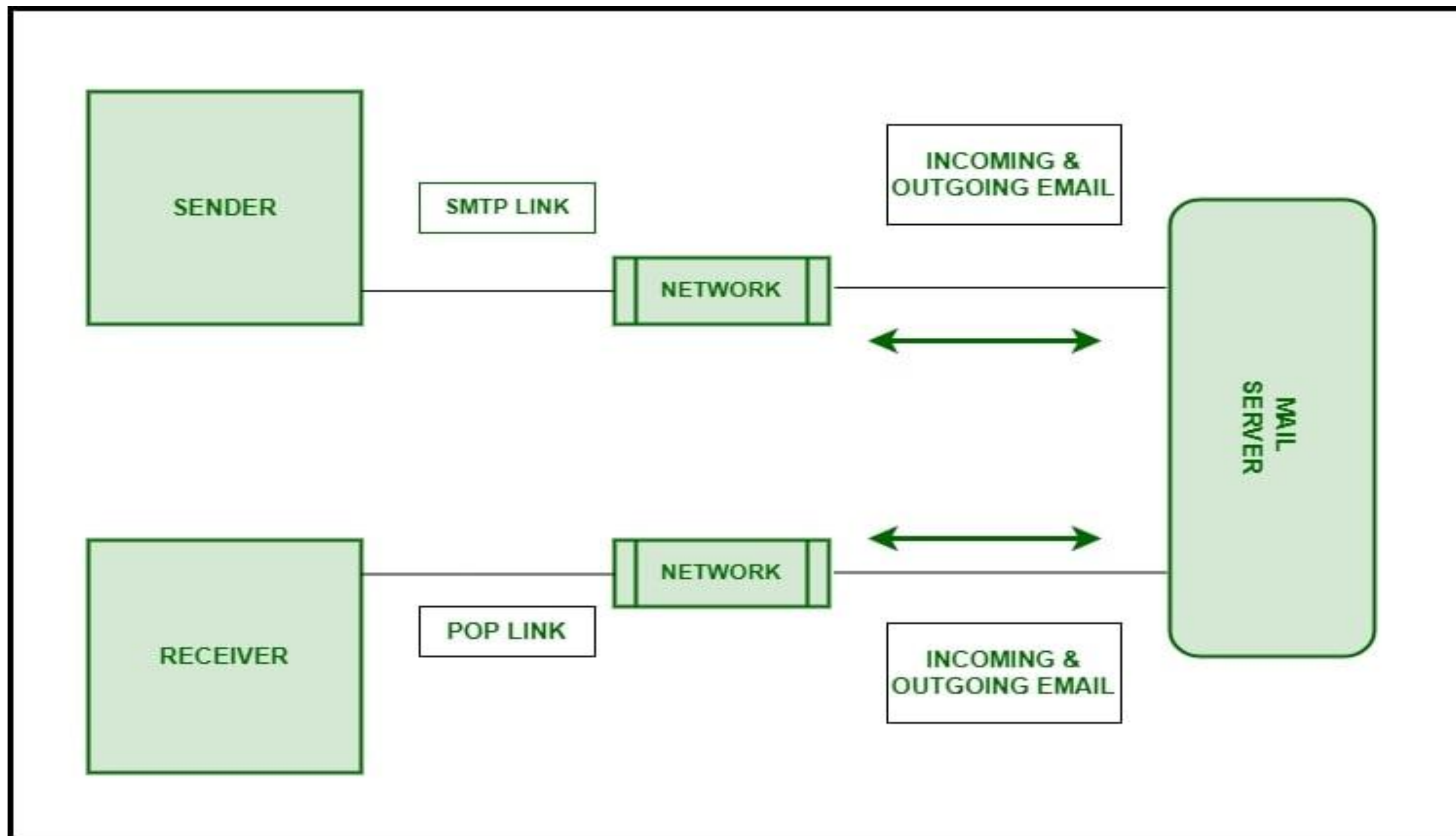


Components of SMTP:

- . **Mail User Agent (MUA):** It is a computer application that helps you in sending and retrieving mail. It is responsible for creating email messages for transfer to the mail transfer agent(MTA).
- . **Mail Submission Agent (MSA):** It is a computer program that receives mail from a Mail User Agent(MUA) and interacts with the Mail Transfer Agent(MTA) for the transfer of the mail.
- . **Mail Transfer Agent (MTA):** It is software that has the work to transfer mail from one system to another with the help of SMTP.
- . **Mail Delivery Agent (MDA):** A mail Delivery agent or Local Delivery Agent is basically a system that helps in the delivery of mail to the local system.

How does SMTP Work?

- . **Communication between the sender and the receiver:** The sender's user agent prepares the message and sends it to the MTA. The MTA's responsibility is to transfer the mail across the network to the receiver's MTA. To send mail, a system must have a client MTA, and to receive mail, a system must have a server MTA.
- . **Sending Emails:** Mail is sent by a series of request and response messages between the client and the server. The message which is sent across consists of a header and a body. A null line is used to terminate the mail header and everything after the null line is considered the body of the message, which is a sequence of ASCII characters. The message body contains the actual information read by the receipt.
- . **Receiving Emails:** The user agent on the server-side checks the mailboxes at a particular time of intervals. If any information is received, it informs the user about the mail. When the user tries to read the mail it displays a list of emails with a short description of each mail.



Advantages of SMTP

- . If necessary, the users can have a dedicated server.
- . It allows for bulk mailing.
- . Low cost and wide coverage area.
- . Offer choices for email tracking.
- . Reliable and prompt email delivery.

Disadvantages of SMTP

- . SMTP's common port can be blocked by several firewalls.
- . SMTP security is a bigger problem.
- . Its simplicity restricts how useful it can be.
- . Just 7-bit ASCII characters can be used.
- . If a message is longer than a certain length, SMTP servers may reject the entire message.

MIME Protocol:

- MIME stands for Multipurpose Internet Mail Extensions. It is used to extend the capabilities of Internet e-mail protocols such as SMTP.
- The MIME protocol allows the users to exchange various types of digital content such as pictures, audio, video, and various types of documents and files in the e-mail.
- MIME was created in 1991 by a computer scientist named Nathan Borenstein at a company called Bell Communications.
- MIME is an e-mail extension protocol, i.e., it does not operate independently, but it helps to extend the capabilities of e-mail in collaboration with other protocols such as SMTP.
- Since MIME was able to transfer only text written file in a limited size English language with the help of the internet. At present, it is used by almost all e-mail related service companies such as Gmail, Yahoo-mail, Hotmail.

Need of MIME Protocol:

MIME protocol is used to transfer e-mail in the computer network for the following reasons:

- 1.The MIME protocol supports multiple languages in e-mail, such as Hindi, French, Japanese, Chinese, etc.
- 2.Simple protocols can reject mail that exceeds a certain size, but there is no word limit in MIME.
- 3.Images, audio, and video cannot be sent using simple e-mail protocols such as SMTP. These require MIME protocol.
- 4.Many times, emails are designed using code such as HTML and CSS, they are mainly used by companies for marketing their product. This type of code uses MIME to send email created from HTML and CSS.

MIME Header:

MIME adds five additional fields to the header portion of the actual e-mail to extend the properties of the simple email protocol. These fields are as follows:

- 1.MIME Version
- 2.Content Type
- 3.Content Type Encoding
- 4.Content Id
- 5.Content description

1. MIME Version

It defines the version of the MIME protocol. This header usually has a parameter value 1.0, indicating that the message is formatted using MIME.

2. Content Type

It describes the type and subtype of information to be sent in the message. These messages can be of many types such as Text, Image, Audio, Video, and they also have many subtypes such that the subtype of the image can be png or jpeg. Similarly, the subtype of Video can be WEBM, MP4 etc.

3. Content Type Encoding

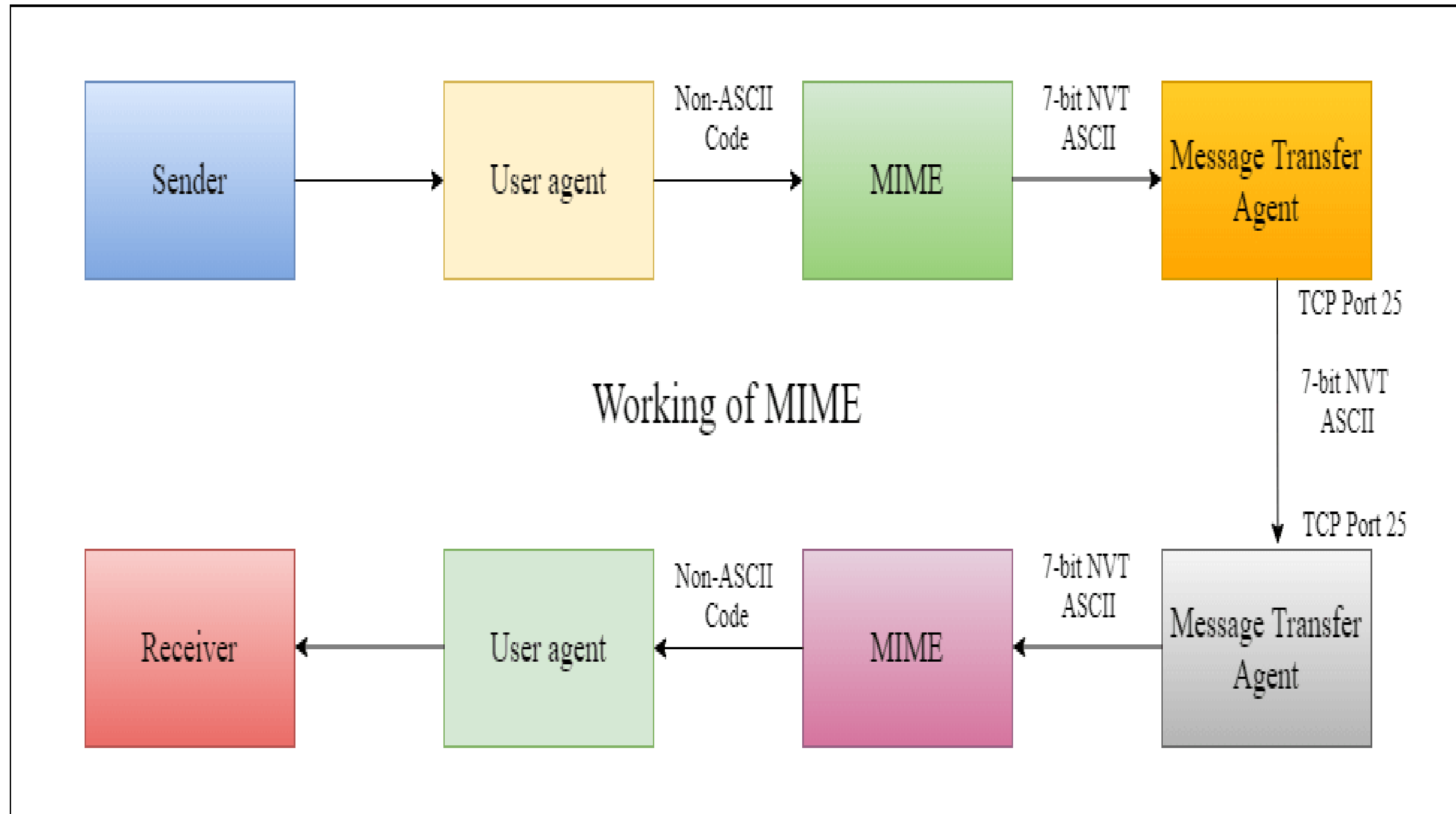
In this field, it is told which method has been used to convert mail information into ASCII or Binary number, such as 7-bit encoding, 8-bit encoding, etc.

4. Content Id

In this field, a unique "Content Id" number is appended to all email messages so that they can be uniquely identified.

5. Content description

This field contains a brief description of the content within the email. This means that information about whatever is being sent in the mail is clearly in the "Content Description". This field also provides the information of name, creation date, and modification date of the file.



Working diagram of MIME Protocol

Features of MIME Protocol

- 1.It supports multiple attachments in a single e-mail.
- 2.It supports the non-ASCII characters.
- 3.It supports unlimited e-mail length.
- 4.It supports multiple languages.

Advantage of the MIME

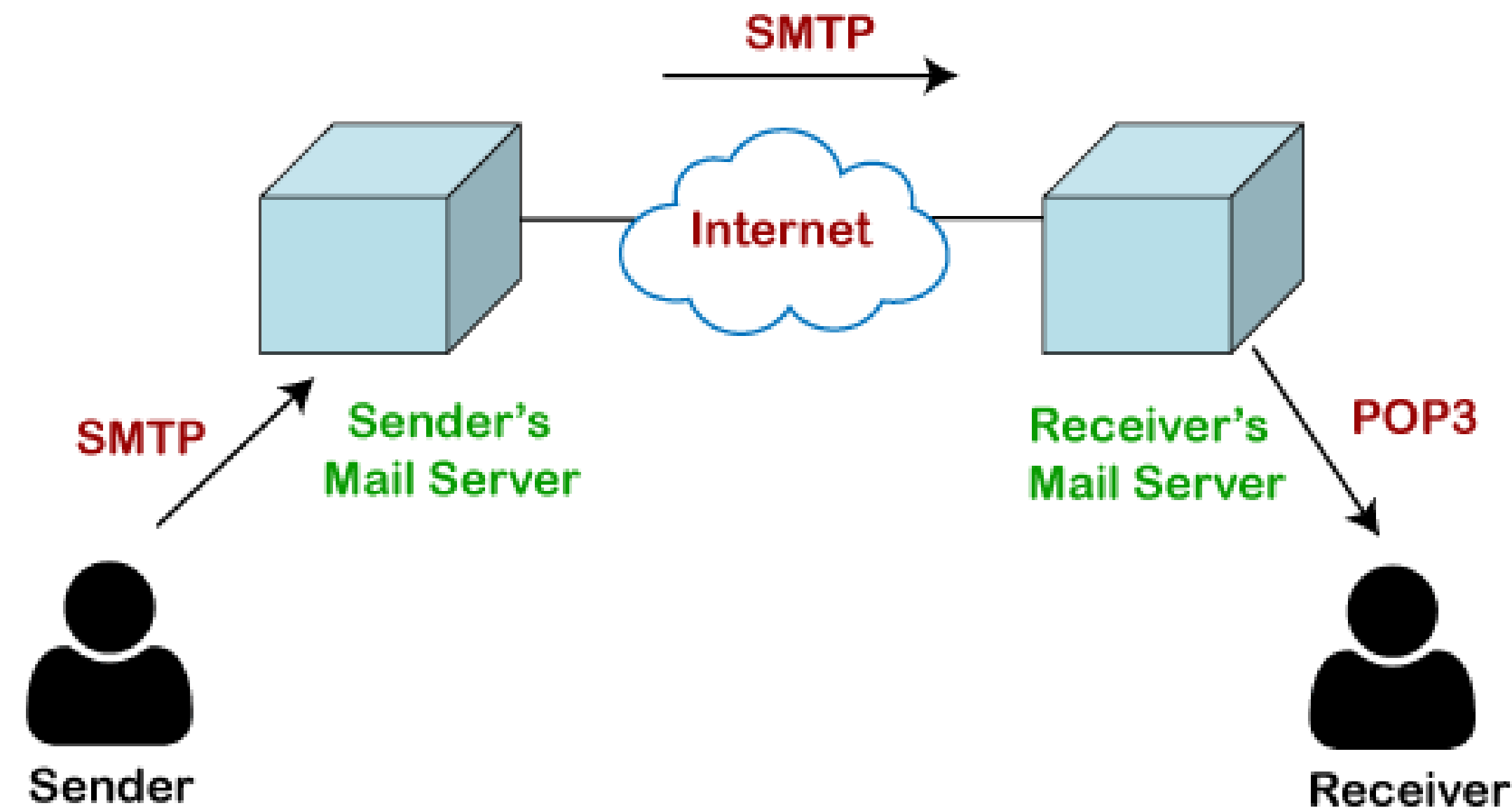
The MIME protocol has the following advantages:

- 1.It is capable of sending various types of files in a message, such as text, audio, video files.
- 2.It also provides the facility to send and receive emails in different languages like Hindi, French, Japanese, Chinese etc.
- 3.It also provides the facility of connecting HTML and CSS to email, due to which people can design email as per their requirement and make it attractive and beautiful.
- 4.It is capable of sending the information contained in an email regardless of its length.
- 5.It assigns a unique id to all e-mails.

POP Protocol:

The POP protocol stands for Post Office Protocol. As we know that SMTP is used as a message transfer agent. When the message is sent, then SMTP is used to deliver the message from the client to the server and then to the recipient server. But the message is sent from the recipient server to the actual server with the help of the Message Access Agent. The Message Access Agent contains two types of protocols, i.e., POP3 and IMAP.

How is mail transmitted?

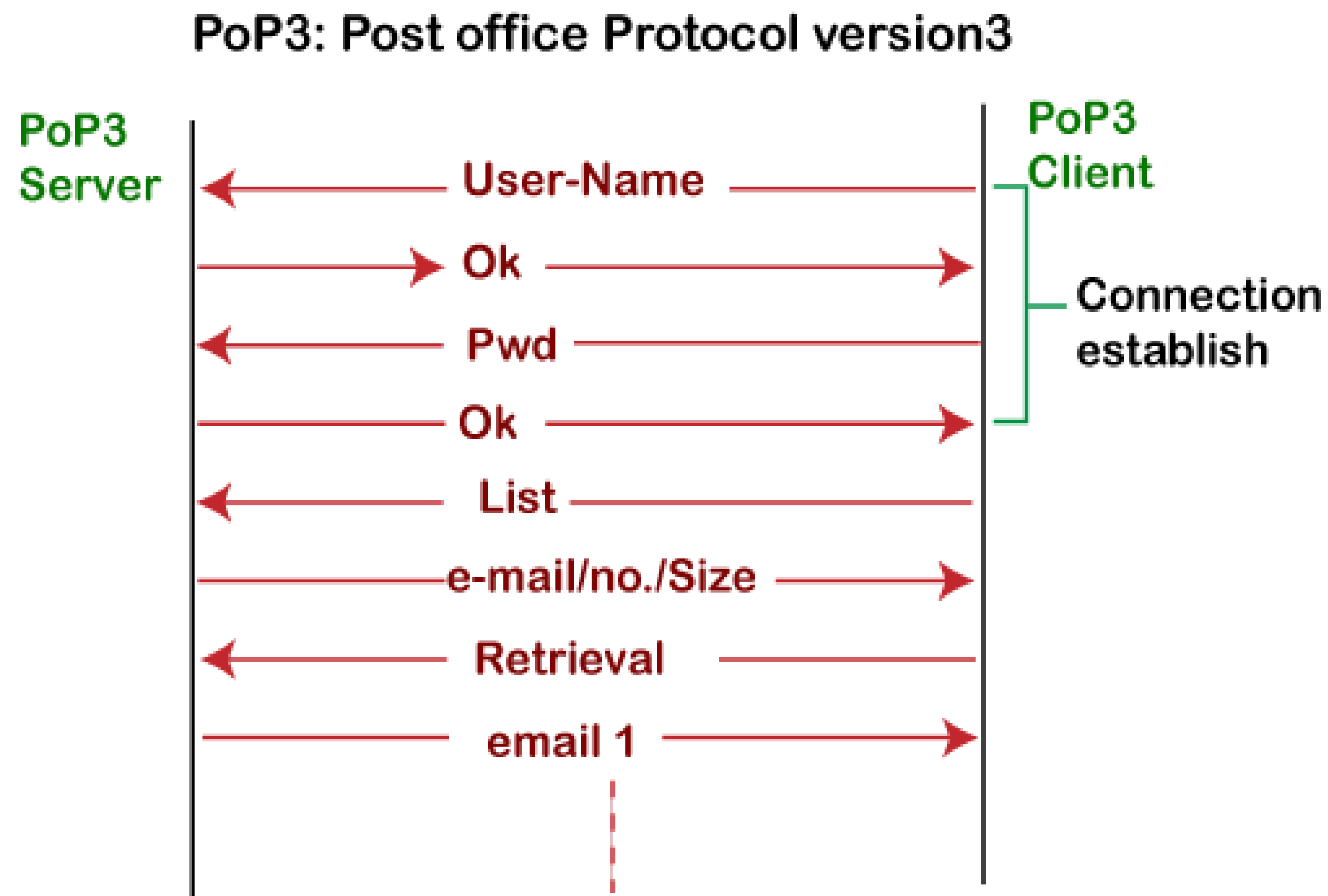


- Suppose sender wants to send the mail to receiver. First mail is transmitted to the sender's mail server. Then, the mail is transmitted from the sender's mail server to the receiver's mail server over the internet. On receiving the mail at the receiver's mail server, the mail is then sent to the user.
- The whole process is done with the help of Email protocols. The transmission of mail from the sender to the sender's mail server and then to the receiver's mail server is done with the help of the SMTP protocol.
- At the receiver's mail server, the POP or IMAP protocol takes the data and transmits to the actual user.
- Since SMTP is a push protocol so it pushes the message from the client to the server.
- As we can observe in the above figure that SMTP pushes the message from the client to the recipient's mail server.
- The third stage of email communication requires a pull protocol, and POP is a pull protocol.
- When the mail is transmitted from the recipient mail server to the client which means that the client is pulling the mail from the server.

What is POP3?

The POP3 is a simple protocol and having very limited functionalities. In the case of the POP3 protocol, the POP3 client is installed on the recipient system while the POP3 server is installed on the recipient's mail server.

Let's understand the working of the POP3 protocol.



- To establish the connection between the POP3 server and the POP3 client, the POP3 server asks for the user name to the POP3 client.
- If the username is found in the POP3 server, then it sends the ok message. It then asks for the password from the POP3 client; then the POP3 client sends the password to the POP3 server.
- If the password is matched, then the POP3 server sends the OK message, and the connection gets established. After the establishment of a connection, the client can see the list of mails on the POP3 mail server.
- In the list of mails, the user will get the email numbers and sizes from the server. Out of this list, the user can start the retrieval of mail.
- Once the client retrieves all the emails from the server, all the emails from the server are deleted. Therefore, we can say that the emails are restricted to a particular machine, so it would not be possible to access the same mails on another machine.
- This situation can be overcome by configuring the email settings to leave a copy of mail on the mail server.

Advantages of POP3 protocol

The following are the advantages of a POP3 protocol:

- It allows the users to read the email offline. It requires an internet connection only at the time of downloading emails from the server. Once the mails are downloaded from the server, then all the downloaded mails reside on our PC or hard disk of our computer, which can be accessed without the internet. Therefore, we can say that the POP3 protocol does not require permanent internet connectivity.
- It provides easy and fast access to the emails as they are already stored on our PC.
- There is no limit on the size of the email which we receive or send.
- It requires less server storage space as all the mails are stored on the local machine.
- There is maximum size on the mailbox, but it is limited by the size of the hard disk.
- It is a simple protocol so it is one of the most popular protocols used today.
- It is easy to configure and use.

Disadvantages of POP3 protocol

The following are the advantages of a POP3 protocol:

- If the emails are downloaded from the server, then all the mails are deleted from the server by default. So, mails cannot be accessed from other machines unless they are configured to leave a copy of the mail on the server.
- Transferring the mail folder from the local machine to another machine can be difficult.
- Since all the attachments are stored on your local machine, there is a high risk of a virus attack if the virus scanner does not scan them. The virus attack can harm the computer.
- The email folder which is downloaded from the mail server can also become corrupted.
- The mails are stored on the local machine, so anyone who sits on your machine can access the email folder.

Webmail

- Webmail is a cloud-based service or Web-based email system that allows you to access and use your email from almost anywhere through an internet connection. Unlike Thunderbird or Microsoft Outlook, it does not need software installation. It is a kind of service, which is provided by certain companies and ISPs (Internet service providers).
- Especially, these kinds of server-based email systems are more popular for younger users. As with Microsoft Outlook, where emails are stored on-site in the hardware storage drive and logging into a connection with the server is needed to get email; so, in this situation, these services provide an appropriate option to email services.

- For people who frequently away from their computers and use multiple devices, Webmail is a great solution for those people. Gmail, Hotmail, and other mainstream providers are the common examples of webmail from Yahoo!, which offer huge amounts of storage, and almost all are free.
- They are very calm to set up and use. Although experts have pointed out, these models have advantages and limitations. With client-side email, users do not need an internet connection to review old emails as they can be archived directly on the computer. However, with webmail, you always need an internet connection to review mails as they are available via the dedicated servers over a network connection.
- Like some resident systems, webmail systems do not need communications protocols; that is one of another benefit of webmail. Some of the less tech-savvy users are frustrated by mail delivery failures while using continue resident or non-webmail systems, but a webmail product helps to prevent that issue.

Some popular webmail services

In modern times, many webmail services are available for users, which are not software-based. Below, a list contains some the free webmail services.

- **Gmail:** Gmail is a type of Webmail, a free Web-based e-mail service that allows users a gigabyte of storage for messages or other data. It is a very popular email service developed by Google. There are 1.5 billion active users of Gmail by October 2019.
- **Yahoo! Mail:** It is a web and cloud-based messaging solution that is launched by the American company Yahoo! on 8 October 1997. You can connect with your email with one-tap access to your inbox with the help of Yahoo! Mail, and it had 225 million users by January 2020. You can use to create Yahoo account by using this link - <https://overview.mail.yahoo.com/>
- **com:** It is a free web-based e-mail service that allows you to send and receive e-mail on your computer. Somewhat, it is like Google's Gmail service but something different in terms of linking desktop Outlook data. Outlook has two types of versions: Microsoft Outlook and Microsoft Outlook Express. To create an Outlook account, you can use this link <https://signup.live.com/?lic=1>

Advantages of Webmail:

Cost

Usually, big providers provide basic webmail services free of cost. Also, some premium services are also offered by some providers.

Large Storage

One of the biggest advantages of webmail is storage capacity. It means the storage in gigabytes or unlimited storage is offered by big webmail providers that allows you to save a large number of messages as well as large attachments. Therefore, freeing up space for other things, you do not require to keep all that data on your own computer.

Disadvantages of Webmail

Although webmail offers users the multiple benefits, it also has some limitations; which are discussed below:

Personal Name

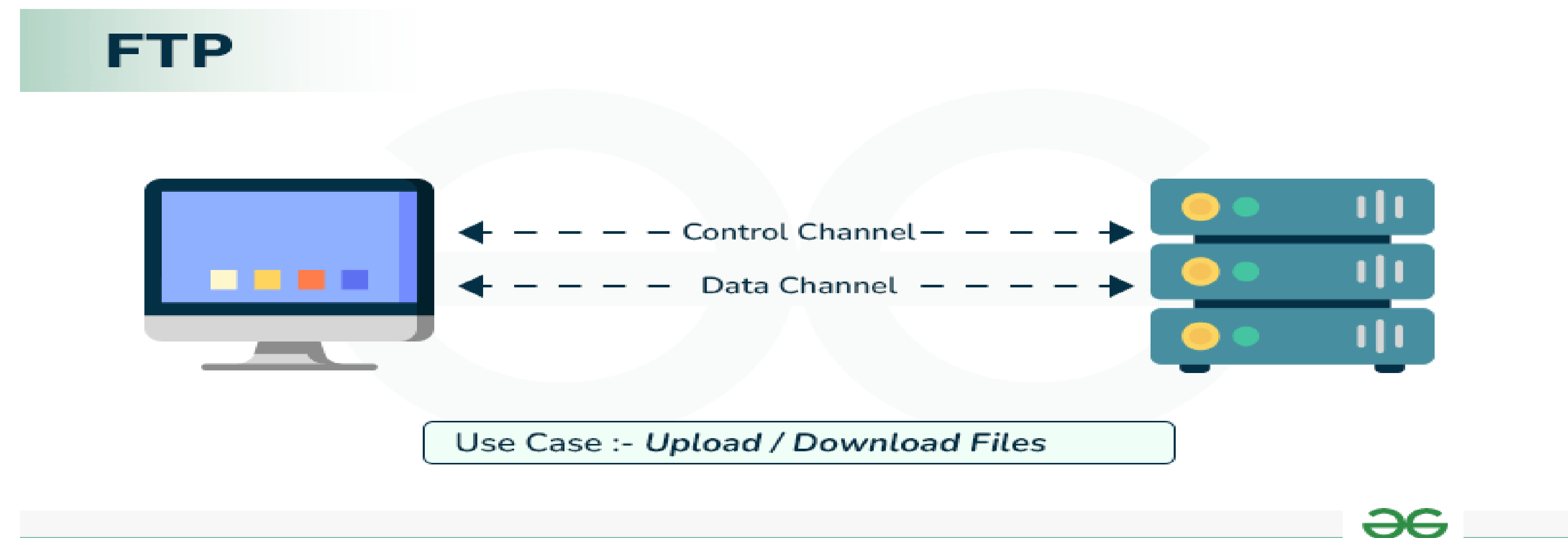
You may be unable to create an address on webmail using the name you had in mind or your own name as it has a large customer base. Whenever you try to create an address on webmail with your own name, it may be this name has taken by another one. Therefore, you cannot address using your own name. You can create an address with a memorable and simple email address that is available. Also, using webmail for business purpose may be a disadvantage because it is considered by many people that the webmail addresses is unprofessional.

Ads

From advertisers, webmail providers have to get their revenue as you are using the service for free of cost. That means you are not allowed to carry personalized advertising, something traditional email services, and even you will have to put up with banners.

FTP:

FTP is a standard communication protocol. There are various other protocols like HTTP which are used to transfer files between computers, but they lack clarity and focus as compared to FTP. Moreover, the systems involved in connection are heterogeneous, i.e. they differ in operating systems, directories, structures, character sets, etc the FTP shields the user from these differences and transfers data efficiently and reliably. FTP can transfer ASCII, EBCDIC, or image files. The ASCII is the default file share format, in this, each character is encoded by NVT ASCII. In ASCII or EBCDIC the destination must be ready to accept files in this mode. The image file format is the default format for transforming binary files.



File Transfer Protocol

Types of FTP

There are different ways through which a server and a client do a file transfer using FTP. Some of them are mentioned below:

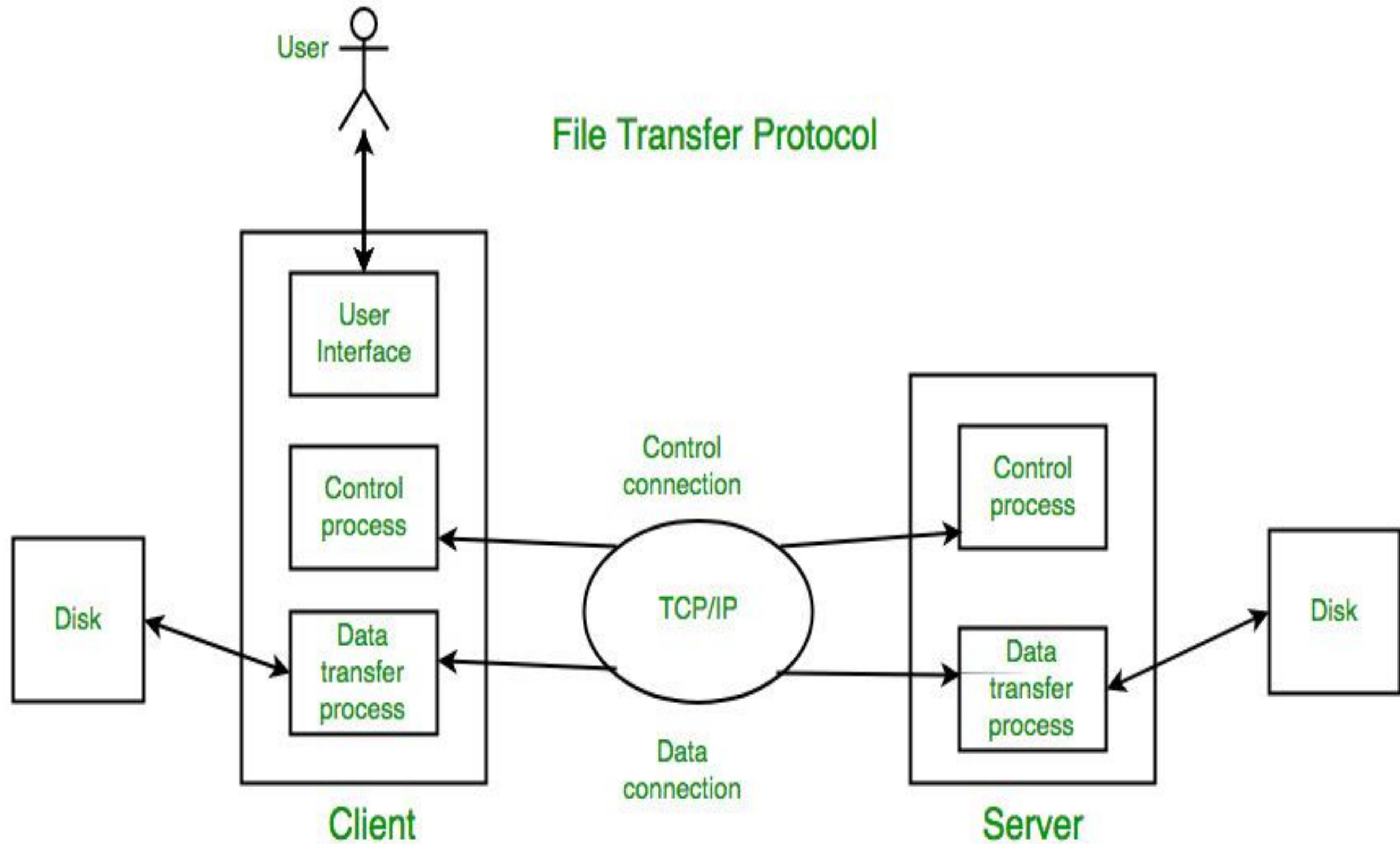
- **Anonymous FTP:** Anonymous FTP is enabled on some sites whose files are available for public access. A user can access these files without having any username or password. Instead, the username is set to anonymous, and the password is to the guest by default. Here, user access is very limited. For example, the user can be allowed to copy the files but not to navigate through directories.
- **Password Protected FTP:** This type of FTP is similar to the previous one, but the change in it is the use of username and password.
- **FTP Secure (FTPS):** It is also called as FTP Secure Sockets Layer (FTP SSL). It is a more secure version of FTP data transfer. Whenever FTP connection is established, Transport Layer Security (TLS) is enabled.
- **Secure FTP (SFTP):** SFTP is not a FTP Protocol, but it is a subset of Secure Shell Protocol, as it works on port 22.

How Does FTP Work:

FTP is a client server protocol that has two communication channel, command channel for conversation control and data channel for file content.

Here are steps mentioned in which FTP works:

- . A user has to log in to FTP Server first, there may be some servers where you can access to content without login, known as anonymous FTP.
- . Client can start a conversation with server, upon requesting to download a file.
- . The user can start different functions like upload, delete, rename, copy files, etc. on server.



Types of Connection in FTP

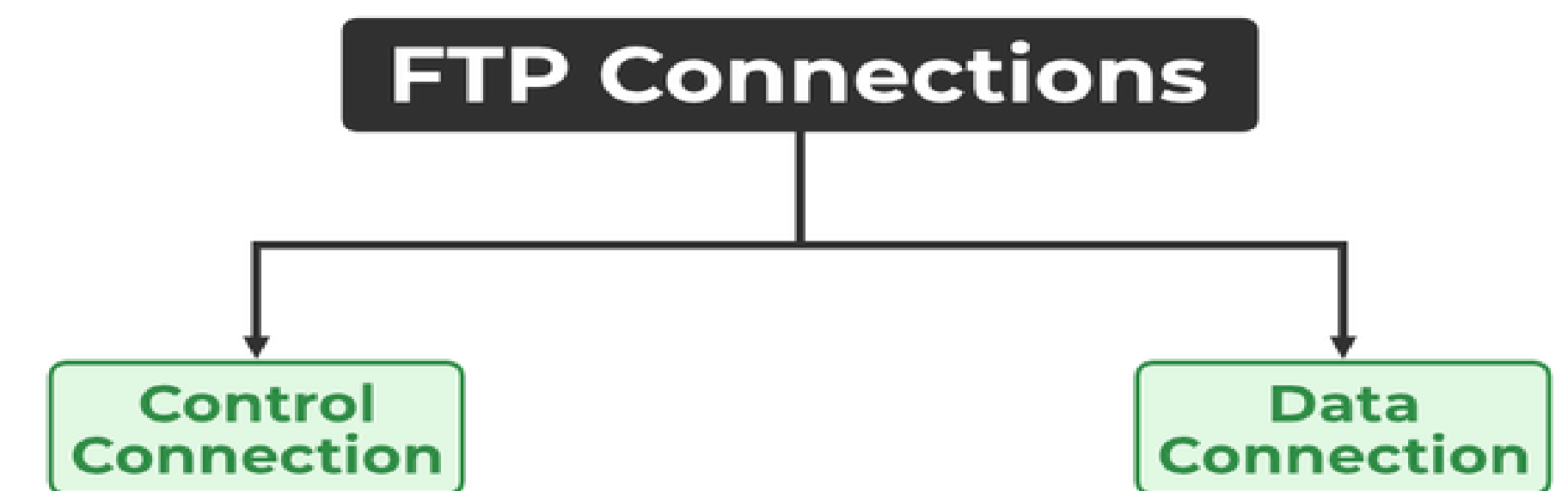
- . Control Connection
- . Data Connection

Control Connection

For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of a control connection. The control connection is initiated on port number 21.

Data connection

For sending the actual file, FTP makes use of a data connection. A data connection is initiated on port number 20. FTP sends the control information out-of-band as it uses a separate control connection. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.



Characteristics of FTP :

- . FTP uses TCP as a transport layer protocol.
- . It is good for simple file transfers, such as during boot time.
- . Errors in the transmission (lost packets, checksum errors) must be handled by the TFTP server.
- . It uses only one connection through well-known port 69.
- . TFTP uses a simple lock-step protocol (each data packet needs to be acknowledged). Thus the throughput is limited.

Advantages of FTP

- . File sharing also comes in the category of advantages of FTP in this between two machines files can be shared on the network.
- . Speed is one of the main benefits of FTP.
- . Since we don't have to finish every operation to obtain the entire file, it is more efficient.
- . Using the username and password, we must log in to the FTP server. As a result, FTP might be considered more secure.
- . We can move the files back and forth via FTP. Let's say you are the firm manager and you provide information to every employee, and they all reply on the same server.

Disadvantages of FTP

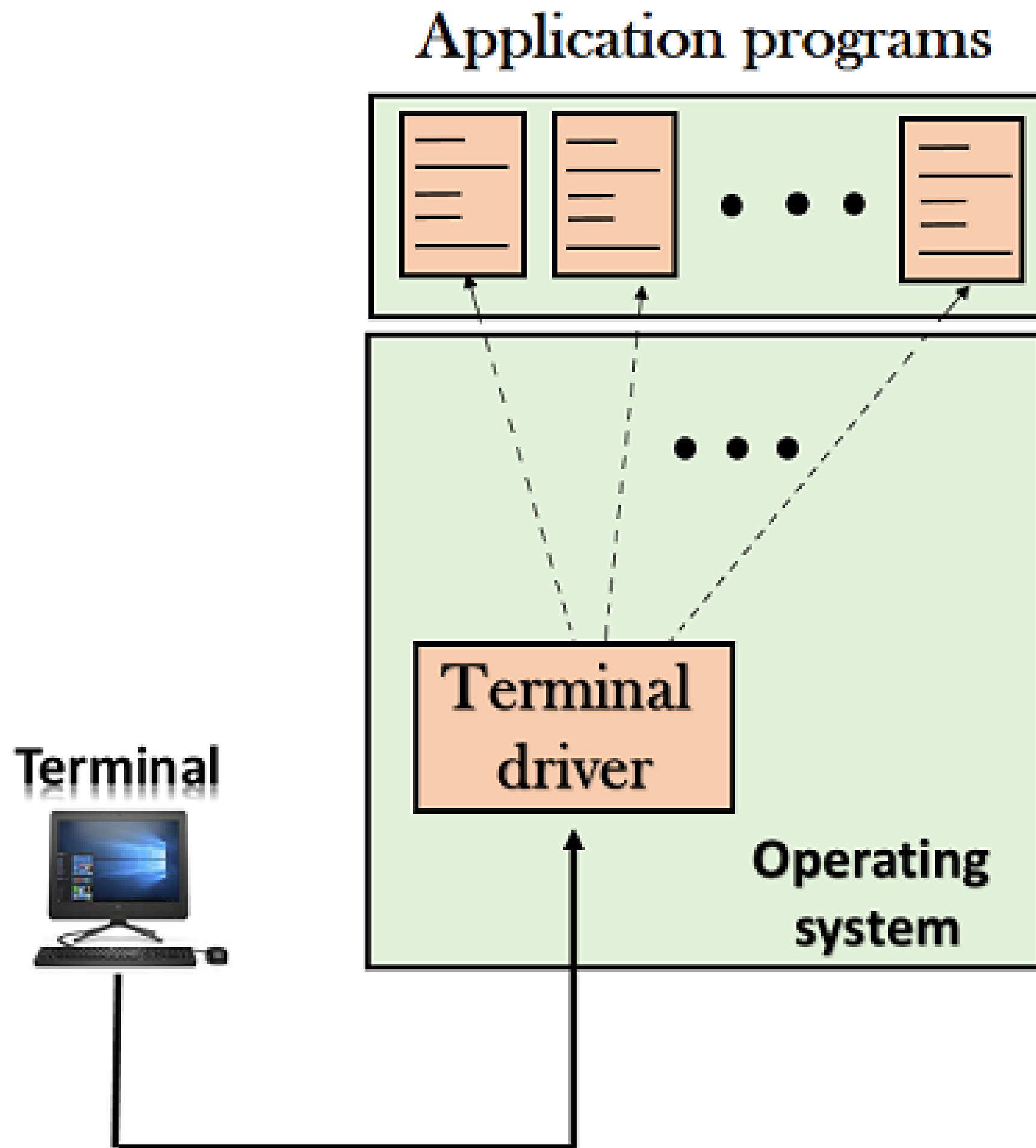
- . File size limit is the drawback of FTP only 2 GB size files can be transferred.
- . More than one receivers are not supported by FTP.
- . FTP does not encrypt the data this is one of the biggest drawbacks of FTP.
- . FTP is unsecured we use login IDs and passwords making it secure but they can be attacked by hackers.

TELNET

- The main task of the internet is to provide services to users. For example, users want to run different application programs at the remote site and transfer a result to the local site. This requires a client-server program such as FTP, SMTP. But this would not allow us to create a specific program for each demand.
- The better solution is to provide a general client-server program that lets the user access any application program on a remote computer. Therefore, a program that allows a user to log on to a remote computer. A popular client-server program Telnet is used to meet such demands. Telnet is an abbreviation for **Terminal Network**.
- Telnet provides a connection to the remote computer in such a way that a local terminal appears to be at the remote side.

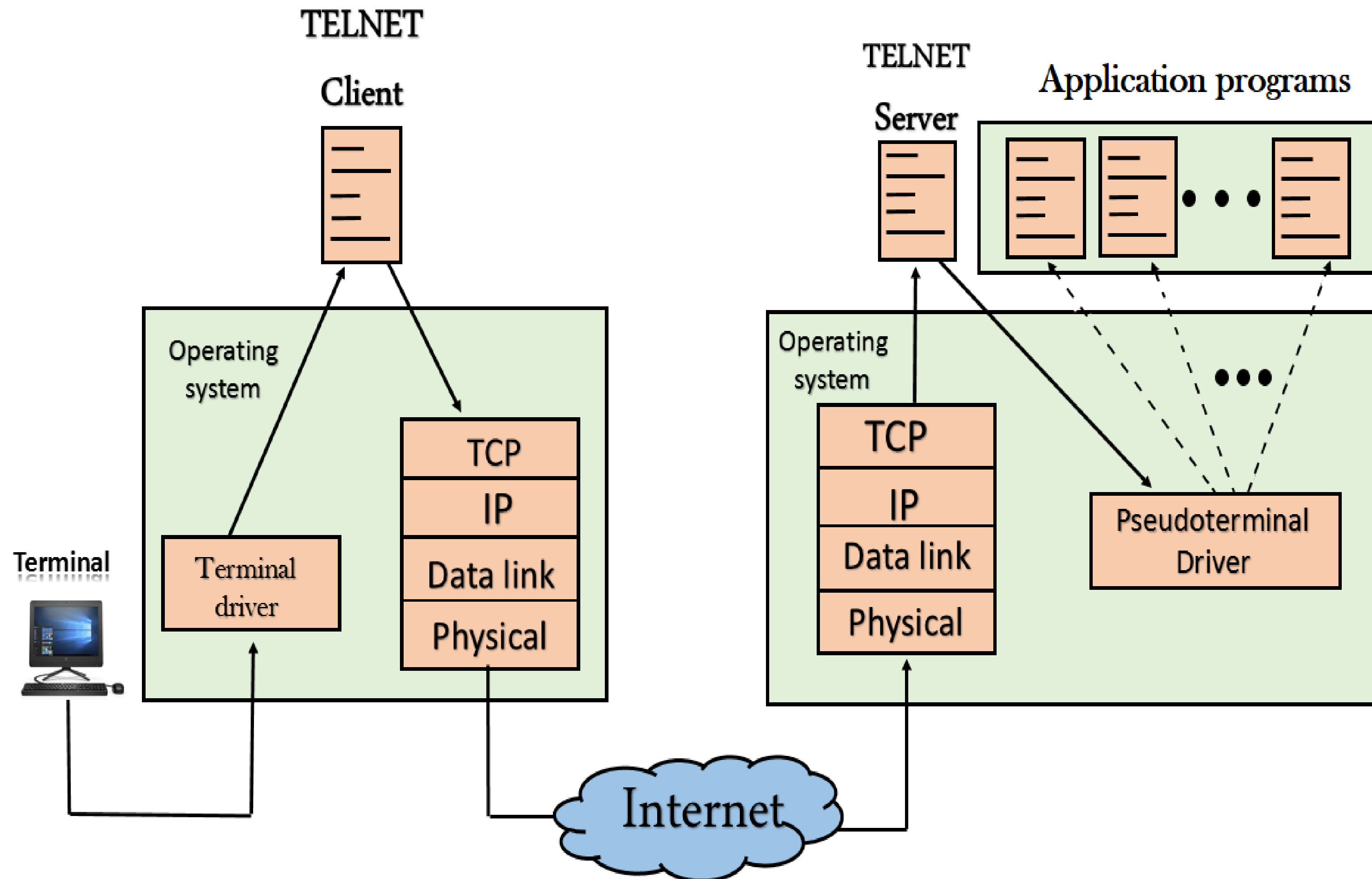
There are two types of login:

Local Login



- When a user logs into a local computer, then it is known as local login.
- When the workstation running terminal emulator, the keystrokes entered by the user are accepted by the terminal driver. The terminal driver then passes these characters to the operating system which in turn, invokes the desired application program.
- However, the operating system has special meaning to special characters. For example, in UNIX some combination of characters has special meanings such as control character with "z" means suspend. Such situations do not create any problem as the terminal driver knows the meaning of such characters. But, it can cause the problems in remote login.

Remote login:



- When the user wants to access an application program on a remote computer, then the user must perform remote login.

How remote login occurs

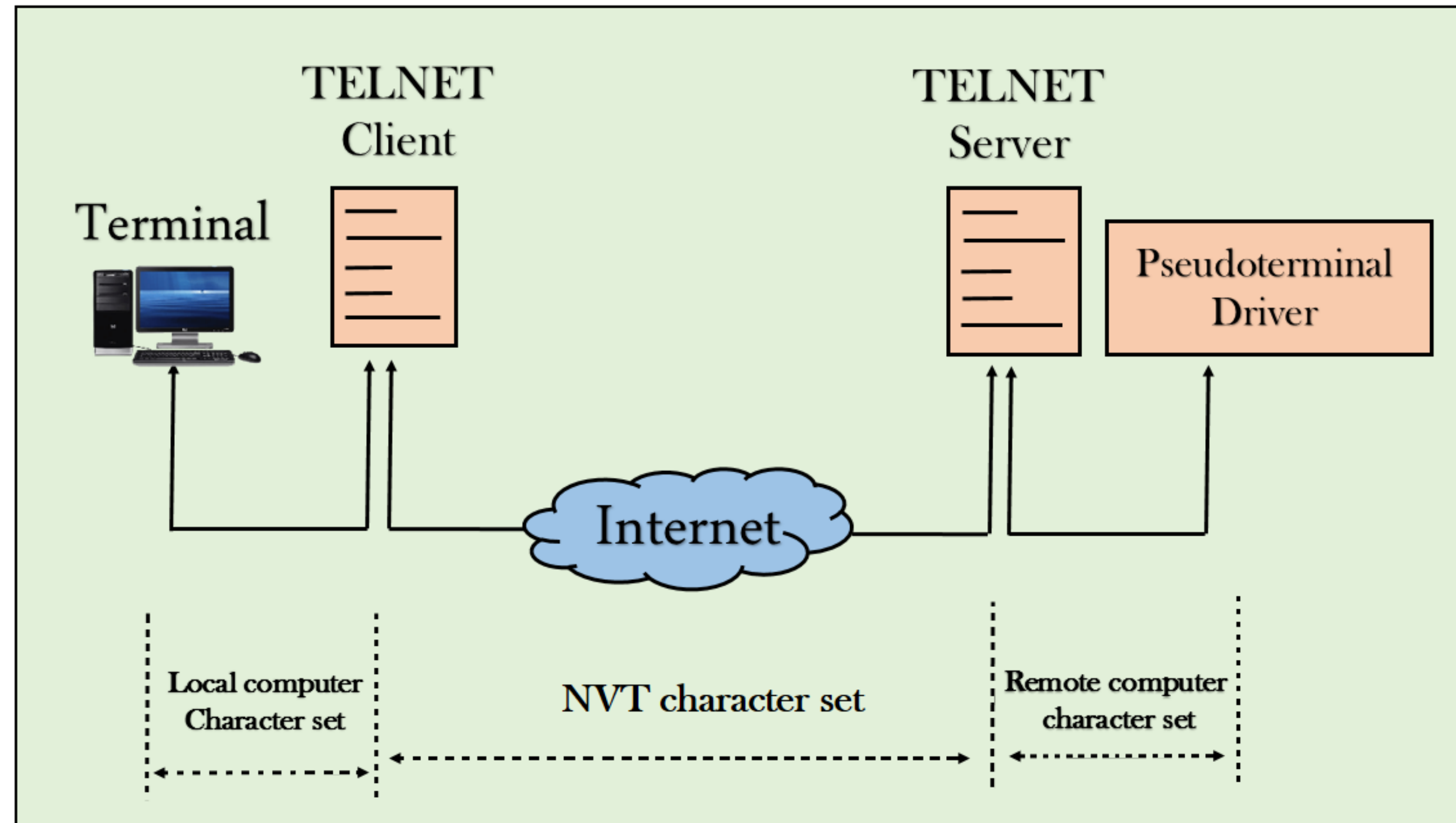
At the local site

The user sends the keystrokes to the terminal driver, the characters are then sent to the TELNET client. The TELNET client which in turn, transforms the characters to a universal character set known as network virtual terminal characters and delivers them to the local TCP/IP stack.

At the remote site

The commands in NVT forms are transmitted to the TCP/IP at the remote machine. Here, the characters are delivered to the operating system and then pass to the TELNET server. The TELNET server transforms the characters which can be understandable by a remote computer. However, the characters cannot be directly passed to the operating system as a remote operating system does not receive the characters from the TELNET server. Therefore it requires some piece of software that can accept the characters from the TELNET server. The operating system then passes these characters to the appropriate application program.

Network Virtual Terminal (NVT)



- The network virtual terminal is an interface that defines how data and commands are sent across the network.
- In today's world, systems are heterogeneous. For example, the operating system accepts a special combination of characters such as end-of-file token running a DOS operating system *ctrl+z* while the token running a UNIX operating system is *ctrl+d*.

- TELNET solves this issue by defining a universal interface known as network virtual interface.
- The TELNET client translates the characters that come from the local terminal into NVT form and then delivers them to the network. The Telnet server then translates the data from NVT form into a form which can be understandable by a remote computer.

Uses of TELNET:

- . Remote Administration and Management
- . Network Diagnostics
- . Understanding Command-Line Interfaces
- . Accessing Bulletin Board Systems (BBS)
- . Automation and Scripting

Advantages of TELNET

- . It provides remote access to someone's computer system.
- . Telnet allows the user for more access with fewer problems in data transmission.
- . Telnet saves a lot of time.
- . The oldest system can be connected to a newer system with telnet having different operating systems.

Disadvantages of TELNET

- . As it is somehow complex, it becomes difficult to beginners in understanding.
- . Data is sent here in form of plain text, that's why it is not so secured.
- . Some capabilities are disabled because of not proper interlinking of the remote and local devices.

Dynamic Host Control Protocol (DHCP):

- Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so they can communicate using IP (Internet Protocol).
- DHCP automates and centrally manages these configurations. There is no need to manually assign IP addresses to new devices. Therefore, there is no requirement for any user configuration to connect to a DHCP based network.
- DHCP can be implemented on local networks as well as large enterprise networks. DHCP is the default protocol used by the most routers and networking equipment. DHCP is also called RFC (Request for comments) 2131.

How DHCP works:

- DHCP runs at the application layer of the TCP/IP protocol stack to dynamically assign IP addresses to DHCP clients/nodes and to allocate TCP/IP configuration information to the DHCP clients.
- Information includes subnet mask information, default gateway, IP addresses and domain name system addresses.
- DHCP is based on client-server protocol in which servers manage a pool of unique IP addresses, as well as information about client configuration parameters, and assign addresses out of those address pools.

The DHCP lease process works as follows:

- First of all, a client (network device) must be connected to the internet.
- DHCP clients request an IP address. Typically, client broadcasts a query for this information.
- DHCP server responds to the client request by providing IP server address and other configuration information. This configuration information also includes time period, called a lease, for which the allocation is valid.
- When refreshing an assignment, a DHCP clients request the same parameters, but the DHCP server may assign a new IP address. This is based on the policies set by the administrator.

Components of DHCP:

When working with DHCP, it is important to understand all of the components. Following are the list of components:

- **DHCP Server:** DHCP server is a networked device running the DHCP service that holds IP addresses and related configuration information. This is typically a server or a router but could be anything that acts as a host, such as an SD-WAN appliance.
- **DHCP client:** DHCP client is the endpoint that receives configuration information from a DHCP server. This can be any device like computer, laptop, IoT endpoint or anything else that requires connectivity to the network. Most of the devices are configured to receive DHCP information by default.
- **IP address pool:** IP address pool is the range of addresses that are available to DHCP clients. IP addresses are typically handed out sequentially from lowest to the highest.

- **Subnet:** Subnet is the partitioned segments of the IP networks. Subnet is used to keep networks manageable.
- **Lease:** Lease is the length of time for which a DHCP client holds the IP address information. When a lease expires, the client has to renew it.
- **DHCP relay:** A host or router that listens for client messages being broadcast on that network and then forwards them to a configured server. The server then sends responses back to the relay agent that passes them along to the client. DHCP relay can be used to centralize DHCP servers instead of having a server on each subnet.

Benefits of DHCP:

There are following benefits of DHCP:

Centralized administration of IP configuration: DHCP IP configuration information can be stored in a single location and enables that administrator to centrally manage all IP address configuration information.

Dynamic host configuration: DHCP automates the host configuration process and eliminates the need to manually configure individual host. When TCP/IP (Transmission control protocol/Internet protocol) is first deployed or when IP infrastructure changes are required.

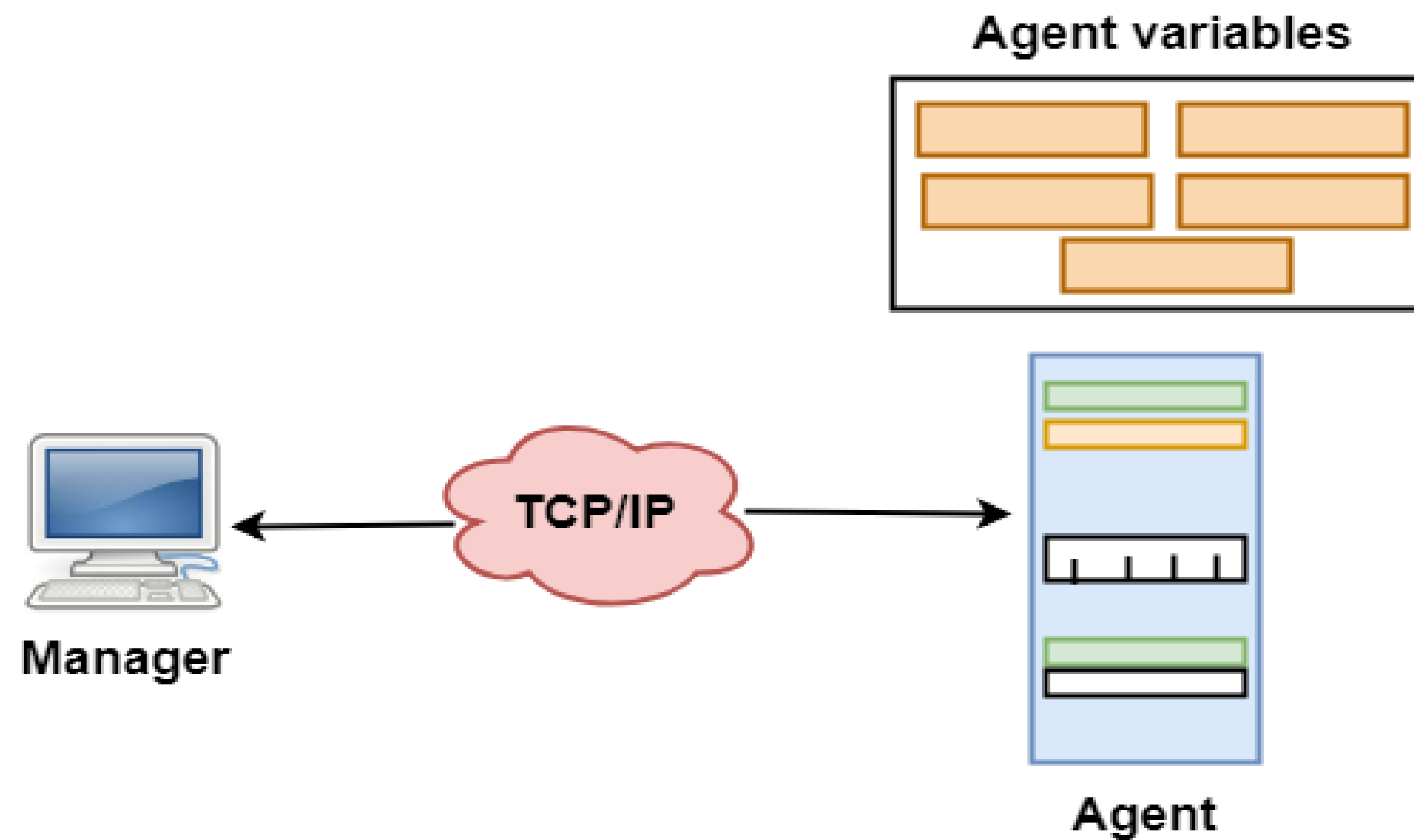
Seamless IP host configuration: The use of DHCP ensures that DHCP clients get accurate and timely IP configuration IP configuration parameter such as IP address, subnet mask, default gateway, IP address of DNS server and so on without user intervention.

Flexibility and scalability: Using DHCP gives the administrator increased flexibility, allowing the administrator to move easily change IP configuration when the infrastructure changes.

Simple Network Management Protocol (SNMP):

- SNMP stands for **Simple Network Management Protocol**.
- SNMP is a framework used for managing devices on the internet.
- It provides a set of operations for monitoring and managing the internet.

SNMP Concept



- SNMP has two components Manager and agent.
- The manager is a host that controls and monitors a set of agents such as routers.
- It is an application layer protocol in which a few manager stations can handle a set of agents.
- The protocol designed at the application level can monitor the devices made by different manufacturers and installed on different physical networks.
- It is used in a heterogeneous network made of different LANs and WANs connected by routers or gateways.

Managers & Agents

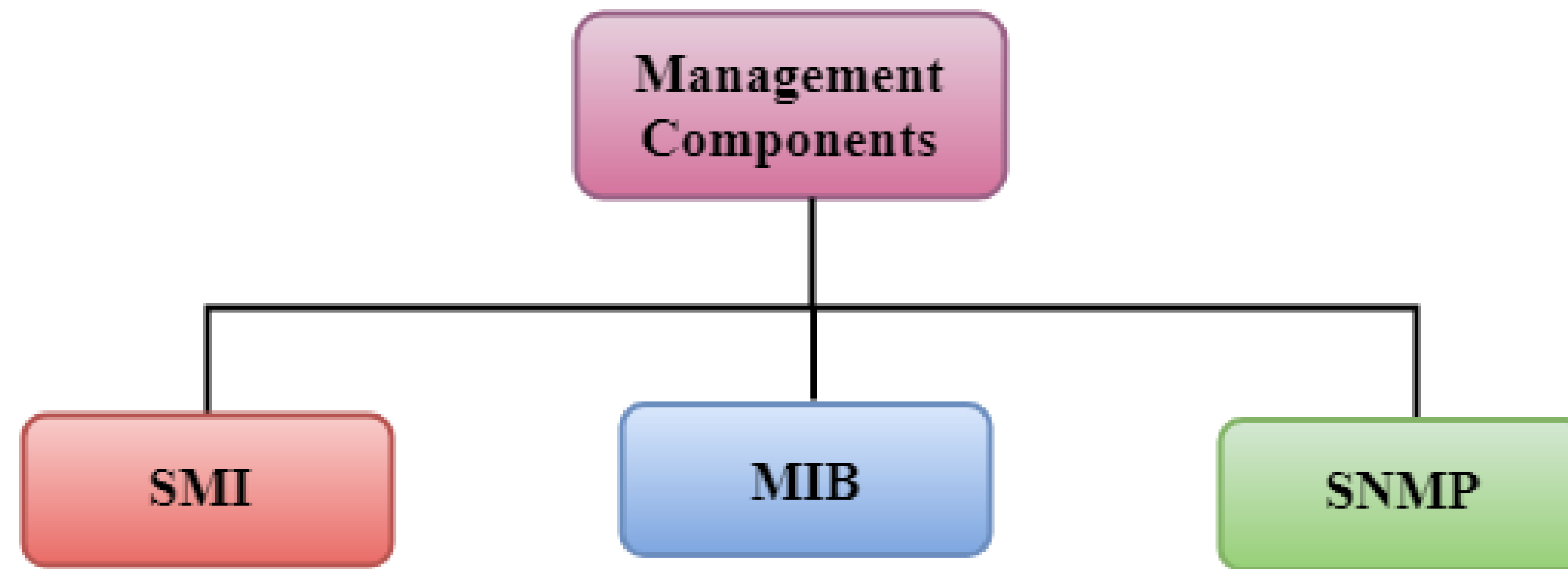
- A manager is a host that runs the SNMP client program while the agent is a router that runs the SNMP server program.
- Management of the internet is achieved through simple interaction between a manager and agent.
- The agent is used to keep the information in a database while the manager is used to access the values in the database. For example, a router can store the appropriate variables such as a number of packets received and forwarded while the manager can compare these variables to determine whether the router is congested or not.
- Agents can also contribute to the management process. A server program on the agent checks the environment, if something goes wrong, the agent sends a warning message to the manager.

Management with SNMP has three basic ideas:

- A manager checks the agent by requesting the information that reflects the behavior of the agent.
- A manager also forces the agent to perform a certain function by resetting values in the agent database.
- An agent also contributes to the management process by warning the manager regarding an unusual condition.

Management Components

- Management is not achieved only through the SNMP protocol but also the use of other protocols that can cooperate with the SNMP protocol. Management is achieved through the use of the other two protocols: SMI (Structure of management information) and MIB(management information base).
- Management is a combination of SMI, MIB, and SNMP. All these three protocols such as abstract syntax notation 1 (ASN.1) and basic encoding rules (BER).

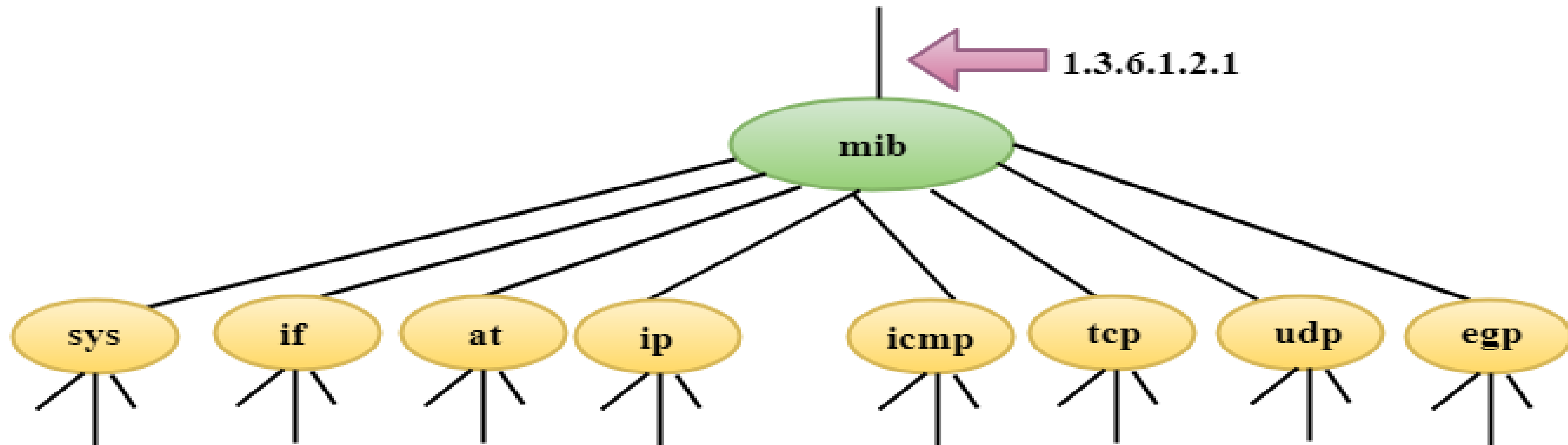


SMI

The SMI (Structure of management information) is a component used in network management. Its main function is to define the type of data that can be stored in an object and to show how to encode the data for the transmission over a network.

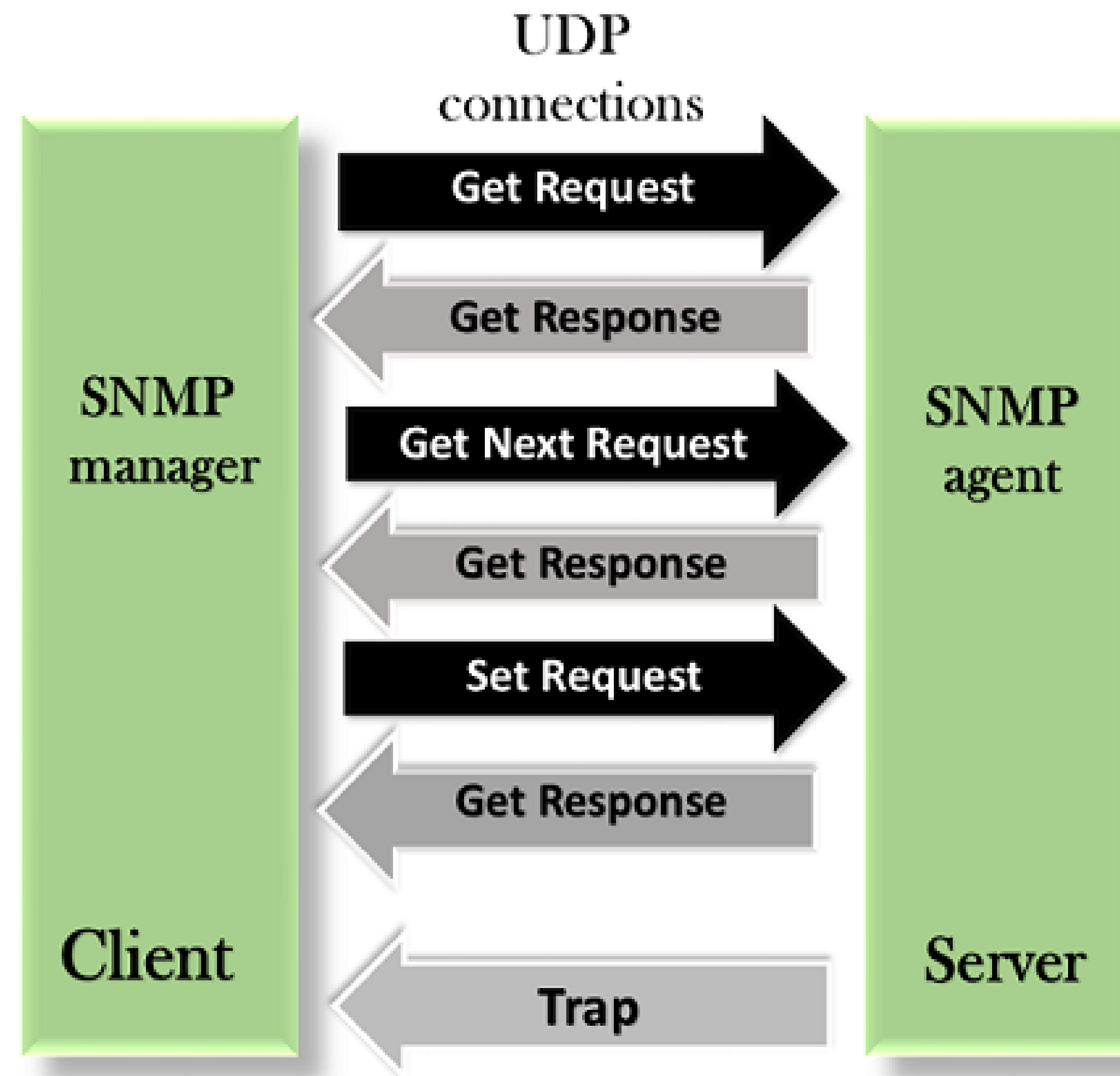
MIB

- The MIB (Management information base) is a second component for the network management.
- Each agent has its own MIB, which is a collection of all the objects that the manager can manage. MIB is categorized into eight groups: system, interface, address translation, ip, icmp, tcp, udp, and egp. These groups are under the mib object.



SNMP

SNMP defines five types of messages: GetRequest, GetNextRequest, SetRequest, GetResponse, and Trap.



GetRequest: The GetRequest message is sent from a manager (client) to the agent (server) to retrieve the value of a variable.

GetNextRequest: The GetNextRequest message is sent from the manager to agent to retrieve the value of a variable. This type of message is used to retrieve the values of the entries in a table. If the manager does not know the indexes of the entries, then it will not be able to retrieve the values. In such situations, GetNextRequest message is used to define an object.

GetResponse: The GetResponse message is sent from an agent to the manager in response to the GetRequest and GetNextRequest message. This message contains the value of a variable requested by the manager.

SetRequest: The SetRequest message is sent from a manager to the agent to set a value in a variable.

Trap: The Trap message is sent from an agent to the manager to report an event. For example, if the agent is rebooted, then it informs the manager as well as sends the time of rebooting.